VESPAW

Theory and methods

Variational electronic structure with projector augmented waves

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Contents

List of symbols iii

1 Introduction 1
- 1.1 Why another look at an old problem . . . 1
- 1.2 VESPAW and CP-PAW . . . 1
- 1.3 How to read this book . . . 2

2 Density-functional theory 3
- 2.1 Introduction . . . 3
- 2.2 The many-electron problem . . . 3
- 2.3 Hohenberg–Kohn theorems . . . 4
- 2.4 Kohn–Sham equations . . . 5
- 2.5 The exchange-correlation energy . . . 5
- 2.6 Exchange-correlation functionals . . . 6
- 2.7 Spin . . . 8
- 2.8 Finite temperature . . . 8

3 Plane waves and Brillouin-zone sampling 11
- 3.1 Introduction . . . 11
- 3.2 Bloch states and plane waves . . . 11
- 3.3 Cutoffs and FFT grids . . . 12
- 3.4 k-point sampling . . . 12
- 3.5 Space-group symmetry . . . 12
- 3.6 The Γ point . . . 13
- 3.7 Slabs, molecules and the dipole correction . . . 13

4 The projector augmented-wave method 15
- 4.1 Introduction . . . 15
- 4.2 The PAW transformation . . . 15
- 4.3 Compensation charges, core and local potential . . . 17
- 4.4 Setup construction . . . 17
- 4.5 Quality checks . . . 17
- 4.6 Relativistic effects . . . 18
- 4.7 The setup library . . . 19
- 4.8 Core-level spectra . . . 19

5 Local hybrid functionals 21
- 5.1 Introduction . . . 21
- 5.2 Hybrid functionals . . . 21
- 5.3 Local orbitals . . . 22
- 5.4 Projection onto the local orbitals . . . 22

5.5 PBE0r . . . 22
5.6 PBE-SL0r . . . 23
5.7 SCAN0r . . . 23
5.8 DFT+U . . . 24

6 Self-consistency: the variational optimizer **25**
6.1 Introduction . . . 25
6.2 The problem with density mixing . . . 25
6.3 Free energy as a function of the Hamiltonian . . . 26
6.4 The gauge direction . . . 26
6.5 Line search and acceptance . . . 27
6.6 Route 2: direct joint minimization . . . 28
6.7 The consistency test . . . 28
6.8 Certificates of a converged state . . . 29

7 Eigensolvers and dynamics **31**
7.1 Introduction . . . 31
7.2 Block Davidson . . . 31
7.3 RMM-DIIS . . . 32
7.4 Conjugate gradients . . . 32
7.5 Car–Parrinello dynamics with friction . . . 32

8 Forces, stress and molecular dynamics **33**
8.1 Introduction . . . 33
8.2 Forces . . . 33
8.3 Stress . . . 34
8.4 Structure relaxation and molecular dynamics . . . 34

9 Structure of the code **36**
9.1 Introduction . . . 36
9.2 Source files . . . 37
9.3 One SCF step . . . 37
9.4 Parallelization . . . 37
9.5 Input . . . 38
9.6 Adding a functional . . . 38
9.7 Adding an optimizer . . . 38
9.8 Regression suite . . . 39

List of symbols

Hartree atomic units are used ($\hbar = m_e = e = 1$, $c \approx 137.036$). Where a symbol has two meanings, the context decides; both are listed. The number of the section or equation that introduces a symbol is given in parentheses.

Electrons and density functionals

ψ_n, ϵ_n, f_n	Kohn–Sham orbital, eigenvalue, occupation
n, $n_\uparrow$, $n_\downarrow$, $\boldsymbol{m}$	density, spin densities, magnetization
τ	kinetic-energy density (meta-GGA)
e_{xc}, v_{xc}, v_τ	XC energy density, XC potential, $\partial e_{\mathrm{xc}}/\partial\tau$
E, A, S	total energy, Mermin free energy $E - TS$, entropy (2.21)
T, μ, N	electronic temperature, Fermi level, number of electrons
$\boldsymbol{\sigma}$	Pauli matrices
a_x	fraction of exact exchange in (local) hybrids
U, J, n_R	Hubbard parameters, on-site occupation matrix (5.8)

Crystal and basis

$\boldsymbol{T}$, $\boldsymbol{b}_i$	lattice vectors, reciprocal lattice vectors
V, $\mathcal{S}$, L_3	cell volume; base area and height of a slab cell (3.7)
$\boldsymbol{k}$, $\boldsymbol{G}$	Bloch vector, reciprocal lattice vector
E_{cut}	plane-wave cutoff (**`EPWPSI`**, in Ry)
$\{\mathrm{M}\|\boldsymbol{t}\}$	space-group operation (3.5)
m, s_0	dipole moment of a slab, position of the potential step (3.7)

PAW method

R, $\boldsymbol{R}$	atom index, atomic position
$\tilde{\psi}_n$	pseudo (PS) orbital, expanded in plane waves
$\mathcal{T}$	PAW transformation, $\psi_n = \mathcal{T}\tilde{\psi}_n$ (4.1)
$\phi_{R,i}$, $\tilde{\phi}_{R,i}$, $\tilde{p}_{R,i}$	AE and PS partial waves, projector functions
$i = (\ell, m, \nu)$	angular momentum and counter of partial waves
r_c	augmentation radius
$D_{R,ij}$	one-center density matrix
$\Delta H_{R,ij}$, $\Delta O_{R,ij}$	one-center Hamiltonian and overlap matrices
$\tilde{H}$, $\hat{O}$, $\tilde{v}$	PS Hamiltonian, overlap operator, PS potential
n^1_R, $\tilde{n}^1_R$	AE and PS one-center densities
$\hat{n}_R$	compensation charge (including nucleus and core)
n^c_R, $\tilde{n}^c_R$	AE core density, pseudized core density
$\bar{v}_R$	local pseudopotential
$M(r)$, d_{rel}	relativistic mass, $1/M - 1$ (4.6)
E_ν	energy at which M is evaluated for the partial waves
$\xi(r)$, $\Xi_{R,\nu\nu'}$	spin–orbit function and its radial integrals (4.6)
χ_a	local orbitals of the hybrid functionals (5.4)

Self-consistency

$\boldsymbol{\theta}$	parameters of the trial Hamiltonian (6.3)
$\mathcal{A}(\boldsymbol{\theta})$	free energy of the state produced by $\boldsymbol{\theta}$
$\boldsymbol{v}_{\text{out}}[\boldsymbol{\theta}]$, $\boldsymbol{r}$	output parameters, residual $\boldsymbol{v}_{\text{out}} - \boldsymbol{\theta}$
χ	static independent-particle response
K, q_0	Kerker preconditioner and its wave vector
$\boldsymbol{s}$	gauge direction (6.4)
β	step length of the line search (6.5)
$\mathcal{A}_{\text{acc}}$, ϵ	last accepted free energy, acceptance slack
Δ_n	frozen-orbital Janak test, $\partial\mathcal{A}/\partial f_n - \epsilon_n$ (6.5)

Forces, stress and relaxation

$\boldsymbol{F}_R$	force on atom R (8.1)
Λ_{mn}	Lagrange multipliers of the orthonormality constraint
$\epsilon_{\alpha\beta}$, $\sigma_{\alpha\beta}$	strain, stress (8.3)
S, P, $\boldsymbol{\sigma}_{\text{ext}}$	internal stress $-\partial E/\partial\epsilon$, target pressure and stress
F, $\boldsymbol{u}_I$	cell deformation, undeformed positions (L-BFGS)
L, B_0, h	cell length scale, bulk modulus, initial curvature (L-BFGS)

Chapter 1

Introduction

1.1 Why another look at an old problem 1
1.2 VESPAW and CP-PAW 1
1.3 How to read this book 2

1.1 Why another look at an old problem

A first-principles calculation of a material ends with a number, a total energy, a band gap, a magnetic moment. Whether this number means anything depends on a long chain of steps: the density functional, the basis set, the treatment of the atomic cores, the sampling of the Brillouin zone, and finally the iteration that makes the electronic structure self-consistent. Most of these steps are controlled by a few numbers that can be converged. The last one is different. A self-consistency cycle that has stalled, or that has settled in a metastable state, may report convergence just like one that has found the ground state, and nothing in the output tells the difference.

For an experienced user who looks at every calculation, this is a nuisance. For the large sets of calculations that are now used to search for new materials for batteries, catalysts and other energy technologies, it is a real problem, because nobody looks at every calculation any more. The materials that matter most, transition-metal oxides with magnetic order, mixed valence and polarons, are exactly the ones where self-consistency is most delicate.

VESPAW starts from this observation. Its central idea is to treat the effective Kohn–Sham Hamiltonian, and not the density, as the variable of the self-consistency problem, and to evaluate the energy for the actual state that each trial Hamiltonian produces. Every energy is then an upper bound of the ground-state energy, and the iteration accepts a step only if this bound does not increase. In this book I describe the physics and the numerical methods on which VESPAW rests, with the emphasis on how the Kohn–Sham equations are solved.

1.2 VESPAW and CP-PAW

VESPAW is a plane-wave code based on the projector augmented-wave (PAW) method [1]. It is a fork of CP-PAW, the Car–Parrinello PAW code written by Peter E. Blöchl at Clausthal University of Technology. The PAW formalism, the construction of the setups, the Car–Parrinello dynamics, the local orbitals and the local hybrid functional PBE0r, and the input format all come from CP-PAW. VESPAW adds

- a *variational* self-consistency optimizer, whose energies are upper bounds of the ground-state energy in every iteration (Chapter 6);
- iterative eigensolvers (block Davidson, RMM-DIIS, conjugate gradients) next to the damped dynamics of CP-PAW (Chapter 7);
- functionals checked against libxc and VASP, the SCAN family (SCAN-L, and SCAN and r^{2}SCAN with the orbital kinetic-energy density), and the local hybrids PBE-SL0r and SCAN0r, which build on PBE0r (Chapters 2 and 5);
- DFT+U, spin–orbit coupling and consistent scalar-relativistic setups (Chapters 4 and 5), the dipole correction for slabs and space-group symmetry (Chapter 3);
- L-BFGS relaxation of atoms and cell (Chapter 8);
- quality checks for PAW setups, an audited setup library (Chapter 4), MPI support for all of the above and a regression suite (Chapter 9).

License. CP-PAW is distributed under the GNU General Public License, version 3 (GPL-3.0). VESPAW is derived from CP-PAW and is distributed under the same license. The older notice "(C) Copyright Clausthal University of Technology, licensed materials" in the headers of the inherited source files predates the GPL release.

1.3 How to read this book

Chapters 2 to 4 summarize the physics that VESPAW implements: density-functional theory, plane waves and the PAW method, including relativistic effects and core-level spectra. Chapter 5 treats hybrid and local hybrid functionals and DFT+U. Chapters 6 and 7 describe how the Kohn–Sham equations are solved, which is where VESPAW differs most from other codes. Chapter 8 covers forces, stress and molecular dynamics, and Chapter 9 the structure of the program. The hands-on tutorial (`doc/tutorial.pdf`) puts all of this into practice.

Input keywords are written in **`bold typewriter`**; their complete description is in the input manual. Hartree atomic units are used throughout unless stated otherwise, while cutoff energies are given in Rydberg, as in the input. The program reports its results in eV, Å and GPa. The symbols used throughout are collected in the list of symbols after the table of contents.

Pitfalls. Many sections end with a short list of pitfalls. They record errors that were actually found in the code and fixed. I have kept them, because they show which quantities are delicate and how they are tested.

Chapter 2

Density-functional theory

2.1 Introduction . . . 3
2.2 The many-electron problem . . . 3
2.3 Hohenberg–Kohn theorems . . . 4
2.4 Kohn–Sham equations . . . 5
2.5 The exchange-correlation energy . . . 5
2.6 Exchange-correlation functionals . . . 6
2.7 Spin . . . 8
2.8 Finite temperature . . . 8

2.1 Introduction

The electrons of a solid interact with each other, and their wave function depends on the coordinates of all of them at once. Already for a few dozen electrons such a wave function cannot even be stored, let alone calculated. Density-functional theory removes this obstacle in two steps. Hohenberg and Kohn showed that the ground-state energy is a functional of the electron density alone, a function of only three coordinates. Kohn and Sham then expressed this functional through a fictitious system of non-interacting electrons that has the same density. All the many-body physics is collected in one term, the exchange-correlation energy, and the art of density-functional theory lies in its approximation. This chapter summarizes the theory and the functionals implemented in VESPAW, with particular attention to how their parameters were checked against independent implementations.

2.2 The many-electron problem

Hamiltonian. A system of nuclei at positions $\boldsymbol{R}_1, \ldots, \boldsymbol{R}_M \equiv \boldsymbol{R}$ with masses M_I and charges Z_I, and of N electrons with coordinates $\boldsymbol{x}_i = (\boldsymbol{r}_i, s_i)$, combining position and spin, is described by the Hamiltonian

$$\hat{H} = \underbrace{-\sum_I \frac{\nabla^2_{\boldsymbol{R}_I}}{2M_I}}_{\hat{T}_\mathrm{N}} \underbrace{-\sum_i \tfrac{1}{2}\nabla^2_{\boldsymbol{r}_i}}_{\hat{T}_\mathrm{e}} + \underbrace{\sum_{i<j} \frac{1}{|\boldsymbol{r}_i - \boldsymbol{r}_j|}}_{\hat{V}_\mathrm{ee}} \underbrace{-\sum_{i,I} \frac{Z_I}{|\boldsymbol{r}_i - \boldsymbol{R}_I|}}_{\hat{V}_\mathrm{eN}} + \underbrace{\sum_{I<J} \frac{Z_I Z_J}{|\boldsymbol{R}_I - \boldsymbol{R}_J|}}_{\hat{V}_\mathrm{NN}} . \tag{2.1}$$

In principle, every property of molecules and solids follows from the eigenstates of this operator. In practice, the number of electronic configurations grows exponentially with the number of electrons, and the problem must be simplified before it can be solved.

Born–Oppenheimer approximation. Nuclei are more than three orders of magnitude heavier than electrons. Born and Oppenheimer [2, 3] therefore first solve the electronic problem for fixed nuclei,

$$\hat{H}_\mathrm{e}(\boldsymbol{R})\,\Phi_k(\boldsymbol{x};\boldsymbol{R}) = E_k(\boldsymbol{R})\,\Phi_k(\boldsymbol{x};\boldsymbol{R}), \qquad \hat{H}_\mathrm{e} = \hat{T}_\mathrm{e} + \hat{V}_\mathrm{ee} + \hat{V}_\mathrm{eN} + \hat{V}_\mathrm{NN}, \tag{2.2}$$

and expand the full wave function as $\Psi(\boldsymbol{x},\boldsymbol{R},t) = \sum_k \Phi_k(\boldsymbol{x};\boldsymbol{R})\chi_k(\boldsymbol{R},t)$. This expansion is exact. Inserting it into the time-dependent Schrödinger equation gives equations of motion for the nuclear wave functions χ_k that are coupled through matrix elements of $\nabla_{\boldsymbol{R}}$ between different electronic states. The Born–Oppenheimer approximation neglects these non-adiabatic couplings: the nuclei then move on a single potential-energy surface $E_k(\boldsymbol{R})$. The approximation fails where two surfaces come close, within the thermal energy $k_B T$ or within the energy of the nuclear motion. Treating the nuclei moreover as classical particles gives Newton's equations

$$M_I \ddot{\boldsymbol{R}}_I = -\nabla_{\boldsymbol{R}_I} E_0(\boldsymbol{R}), \tag{2.3}$$

which is the basis of molecular dynamics (Chapter 8). By the Hellmann–Feynman theorem [4] the force is the expectation value $-\langle\Phi_0|\nabla_{\boldsymbol{R}_I}\hat{H}_\mathrm{e}|\Phi_0\rangle$. What remains is to find the electronic ground state Φ_0 and its energy $E_0(\boldsymbol{R})$.

Variational principle. For any normalizable, antisymmetric trial wave function Φ,

$$E[\Phi] = \frac{\langle\Phi|\hat{H}_\mathrm{e}|\Phi\rangle}{\langle\Phi|\Phi\rangle} \geq E_0, \tag{2.4}$$

with equality only for the ground state. Wave-function methods restrict Φ to a subspace: Hartree–Fock theory to a single Slater determinant, configuration interaction to linear combinations of determinants [5]. Their accuracy can be improved systematically by enlarging the subspace, but the cost grows so steeply that only small systems can be treated. Density-functional theory avoids this cost by replacing the many-electron wave function with the density. The upper-bound property of Eq. (2.4) carries over to it, and it is the foundation of the variational optimizer of Chapter 6.

2.3 Hohenberg–Kohn theorems

The electron density

$$n(\boldsymbol{r}) = N \sum_s \int d^4x_2 \cdots \int d^4x_N\, |\Phi(\boldsymbol{r}s, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_N)|^2, \qquad \int d^4x := \sum_s \int d^3r, \tag{2.5}$$

depends on three coordinates only. Hohenberg and Kohn [6] showed that it nevertheless determines all ground-state properties.

First theorem. The external potential is determined, up to a constant, by the ground-state density. Suppose that two potentials v_1 and v_2, which differ by more than a constant, lead to non-degenerate ground states Φ_1 and Φ_2 with the same density n. Because Φ_2 is not the ground state of $\hat{H}_1$,

$$E_1 < \langle\Phi_2|\hat{H}_1|\Phi_2\rangle = \langle\Phi_2|\hat{H}_2|\Phi_2\rangle + \langle\Phi_2|\hat{H}_1 - \hat{H}_2|\Phi_2\rangle = E_2 + \int d^3r\, n(\boldsymbol{r})\big[v_1(\boldsymbol{r}) - v_2(\boldsymbol{r})\big]. \tag{2.6}$$

Exchanging the indices gives $E_2 < E_1 + \int d^3r\, n\,(v_2 - v_1)$, and the sum of both inequalities gives the contradiction $E_1 + E_2 < E_1 + E_2$. Since v fixes the Hamiltonian, the density fixes the ground state and all properties derived from it.

Second theorem. The energy can be written as

$$E_v[n] = F[n] + \int d^3r\, v(\boldsymbol{r})\, n(\boldsymbol{r}), \qquad F[n] = \min_{\Phi\to n} \langle \Phi | \hat{T}_\mathrm{e} + \hat{V}_\mathrm{ee} | \Phi \rangle, \tag{2.7}$$

where F is a universal functional, the same for all systems, and the minimum runs over all wave functions with density n (constrained search of Levy [7], which also removes the restriction to non-degenerate ground states). By Eq. (2.4), $E_v[n]$ is minimized by the ground-state density, and its minimum is the ground-state energy. If F were known, the many-electron problem would be solved for all external potentials at once. The theorems hold for the ground state only; excited states require extensions such as time-dependent density-functional theory.

2.4 Kohn–Sham equations

The kinetic energy is the largest part of $F[n]$ and the hardest to express directly by the density. Kohn and Sham [8] therefore introduce a fictitious system of non-interacting electrons that has the same density as the interacting one. Its kinetic energy T_s is calculated exactly from its orbitals, and only the remainder is approximated. With non-interacting orbitals ψ_n with occupations f_n,

$$E[n] = \sum_n f_n \langle \psi_n | -\tfrac{1}{2}\nabla^2 | \psi_n \rangle + \int d^3r\, v_\mathrm{ext} n + \frac{1}{2} \iint d^3r\, d^3r' \, \frac{n(\boldsymbol{r}) n(\boldsymbol{r}')}{|\boldsymbol{r} - \boldsymbol{r}'|} + E_\mathrm{xc}[n], \qquad n(\boldsymbol{r}) = \sum_n f_n |\psi_n(\boldsymbol{r})|^2, \tag{2.8}$$

the terms are the kinetic energy of the non-interacting electrons, the interaction with the external potential of the nuclei, the Hartree energy, and the exchange-correlation energy, which contains everything else. The nuclear repulsion V_NN is a constant for fixed nuclei and is added to E; it matters for forces and stress. Minimizing with orthonormal orbitals gives the Kohn–Sham equations

$$\Big[-\tfrac{1}{2}\nabla^2 + v_\mathrm{eff}(\boldsymbol{r}) \Big] \psi_n = \epsilon_n \psi_n, \qquad v_\mathrm{eff} = v_\mathrm{ext} + v_\mathrm{H}[n] + \frac{\delta E_\mathrm{xc}}{\delta n}. \tag{2.9}$$

Because v_eff depends on the density, the equations must be solved self-consistently (Chapter 6). A self-consistent solution is a stationary point of E only if the potential is the functional derivative of the energy. This condition is easy to state and easy to violate in an implementation; VESPAW tests it routinely (Sec. 6.7).

2.5 The exchange-correlation energy

Exchange-correlation hole. If the electrons were uncorrelated, the probability of finding one electron at $\boldsymbol{r}$ and another at $\boldsymbol{r}'$ would be the product $n(\boldsymbol{r})n(\boldsymbol{r}')$, and their interaction would be the Hartree energy. Real electrons avoid each other, because of the Pauli principle (exchange) and because of their Coulomb repulsion (correlation). The pair density is therefore written as

$$n^{(2)}(\boldsymbol{r}, \boldsymbol{r}') = n(\boldsymbol{r}) \big[n(\boldsymbol{r}') + h(\boldsymbol{r}, \boldsymbol{r}') \big], \tag{2.10}$$

where the exchange-correlation hole $h(\boldsymbol{r}, \boldsymbol{r}')$ describes the depletion of density around an electron at $\boldsymbol{r}$. It obeys the sum rule

$$\int d^3r'\, h(\boldsymbol{r}, \boldsymbol{r}') = -1 : \tag{2.11}$$

an electron sees exactly $N - 1$ others. The electron–electron interaction becomes the Hartree energy plus the interaction of each electron with its own hole,

$$V_\mathrm{ee} = E_\mathrm{H} + U_\mathrm{xc}, \qquad U_\mathrm{xc} = \frac{1}{2} \int d^3r\, n(\boldsymbol{r}) \int d^3r' \, \frac{h(\boldsymbol{r}, \boldsymbol{r}')}{|\boldsymbol{r} - \boldsymbol{r}'|}. \tag{2.12}$$

U_xc is not yet E_xc. The exchange-correlation energy also contains the difference $T - T_s$ between the kinetic energy of the interacting electrons and that of the Kohn–Sham system.

Adiabatic connection. Both contributions are combined by switching on the interaction, $\hat{V}_{ee} \to \lambda \hat{V}_{ee}$, while an external potential keeps the density fixed [9–11]. At $\lambda = 0$ one has the Kohn–Sham system, at $\lambda = 1$ the real one. Integrating over λ gives

$$E_{xc} = \int_0^1 d\lambda\, U_{xc}^{\lambda} = \frac{1}{2} \int d^3r\, n(\boldsymbol{r}) \int d^3r'\, \frac{\bar{h}(\boldsymbol{r}, \boldsymbol{r}')}{|\boldsymbol{r} - \boldsymbol{r}'|}, \qquad \bar{h} = \int_0^1 d\lambda\, h^{\lambda}, \tag{2.13}$$

so the exchange-correlation energy is the electrostatic energy of the electrons with their coupling-constant-averaged hole; the kinetic part is absorbed in $\bar{h}$. At $\lambda = 0$ the hole is the exchange hole of the Kohn–Sham determinant, and U_{xc}^0 is the exact (Hartree–Fock-like) exchange energy E_x evaluated with the Kohn–Sham orbitals. Eq. (2.13) is the starting point of the hybrid functionals (Chapter 5). The sum rule explains why simple approximations of $\bar{h}$ work better than one might expect: only the spherical average of the hole enters the energy, and its normalization is fixed.

Self-interaction. For a single electron, with density $n_\uparrow = n$, $n_\downarrow = 0$, there is no electron–electron interaction. The exact functional must then satisfy

$$E_{xc}[n, 0] = -E_H[n], \qquad E_c[n, 0] = 0 : \tag{2.14}$$

exchange cancels the spurious interaction of the electron with itself in the Hartree term. Semilocal functionals violate this condition. The self-interaction error $E_H[n] + E_{xc}[n, 0]$ delocalizes electrons, in particular the d and f electrons of transition-metal and rare-earth compounds, and reduces band gaps. Exact exchange is free of self-interaction; this is the motivation for the hybrid functionals of Chapter 5.

2.6 Exchange-correlation functionals

LDA. The local-density approximation, proposed by Kohn and Sham [8], takes at each point the exchange-correlation energy per electron ϵ_{xc}^{hom} of the homogeneous electron gas with the local density,

$$E_{xc}^{LDA} = \int d^3r\, n(\boldsymbol{r})\, \epsilon_{xc}^{hom}\big(n(\boldsymbol{r})\big). \tag{2.15}$$

Exchange is known analytically; correlation is parametrized from the quantum Monte Carlo data of Ceperley and Alder [12] (PZ [13], PW92 [14]). The LDA works well for many solids, but it overbinds, severely so for molecules.

GGA. Generalized gradient approximations [15] multiply the LDA energy density by an enhancement factor that depends on the density and its gradient,

$$E_{xc}^{GGA} = \int d^3r\, n\, \epsilon_x^{hom}(n)\, F_{xc}(r_s, s), \qquad s = \frac{|\nabla n|}{2k_F n}, \qquad k_F = (3\pi^2 n)^{1/3}, \tag{2.16}$$

with the Seitz radius $r_s = (3/4\pi n)^{1/3}$ and the reduced gradient s. GGAs are not simply the next order of a gradient expansion, which is poor for real densities; they are constructed to satisfy exact conditions such as the sum rule of the hole. They correct most of the overbinding of the LDA and made density-functional theory useful in chemistry. The default in VESPAW is PBE [16].

Selection. All functionals are selected by **`!CONTROL!DFT TYPE`**, as a number or a name, e.g. **`TYPE='PBE'`**; the VASP **`GGA`** tags are accepted as names. Table 2.1 lists them.

Parameters. The constants follow the original publications and agree with libxc and VASP: $\beta = 0.06672455060314922$ in PBE correlation and in $\mu = \beta\pi^2/3$ of PBE exchange (before October 2026 VESPAW used $\beta = \nu C_c(0) = 0.0667263$, kept by **`TYPE=-10`** together with the pre-2024 reduced gradient, t^2 too small by a factor 0.62); PW91 keeps $\beta = \nu C_c(0)$, as in VASP; PBEsol uses $\mu = 10/81$ and $\beta = 0.046$; SCAN-L uses the optimized Perdew–Constantin parameters $a = 1.784720$, $b = 0.258304$ of Ref. [26]; the spin interpolation of SCAN correlation uses $G_c(\zeta) = [1 - 2.363(d_x(\zeta) - 1)](1 - \zeta^{12})$, with 2.363 as in VASP and libxc (the SCAN supplement quotes 2.3631). Unit conversions use CODATA 2018 (1 Hartree = 27.211386 eV, 1 bohr = 0.529177211 Å). Plane-wave hybrids with full exact exchange (PBE0, HSE06, SCAN0, B3LYP) are not yet available.

Meta-GGAs: SCAN, r^{2}SCAN, SCAN-L and TSCAN. The SCAN functional [24] depends in addition on the kinetic energy density $\tau = \frac{1}{2}\sum_n f_n|\nabla\psi_n|^2$ through the iso-orbital indicator

$$\alpha = \frac{\tau - \tau_W}{\tau_{\text{unif}}}, \qquad \tau_W = \frac{|\nabla n|^2}{8n}, \qquad \tau_{\text{unif}} = \tfrac{3}{10}(3\pi^2)^{2/3} n^{5/3}. \tag{2.17}$$

VESPAW provides two routes:

- **SCAN-L** (**`TYPE='SCANL'`**), the deorbitalized SCAN of Mejía-Rodríguez and Trickey [26, 28], in which α is approximated from n, $|\nabla n|$ and $\nabla^2 n$. The functional depends on the Laplacian of the density, so its potential contains second-derivative terms (see below).
- **TSCAN** (**`TYPE='SCAN'`** with **`TSCANDELOC=T`**), the orbital-dependent SCAN with the true τ. The Hamiltonian then contains the operator $-\frac{1}{2}\nabla \cdot v_\tau \nabla$ with $v_\tau = \partial e_{\text{xc}}/\partial\tau$. In the PAW spheres the one-center τ is built from the partial waves, including its non-spherical parts.
- **r^{2}SCAN** (**`TYPE='R2SCAN'`** with **`TSCANDELOC=T`**), the regularized and restored SCAN of Furness *et al.* [25], with the same orbital τ and the same PAW treatment as TSCAN. It replaces α by the regularized $\bar\alpha = (\tau - \tau_W)/(\tau_{\text{unif}} + \eta\tau_W)$ with $\eta = 10^{-3}$, the switching function of SCAN by a polynomial for $0 \le \bar\alpha \le 2.5$, and adds the gradient-expansion corrections of exchange and correlation. The implementation follows the paper and its erratum and agrees with libxc to 10^{-14}. Band gaps change smoothly with the cutoff, unlike those of SCAN, which scatter by a few 0.01 eV.

PAW treatment of τ. The kinetic-energy density is divided like the density: the plane-wave part $\tilde\tau$ of the auxiliary wave functions plus the pseudized core kinetic-energy density $\tilde\tau_c$, and in each sphere the one-center terms $\tau^1 - \tilde\tau^1 = \sum_{ij} D_{ij}\left(\frac{1}{2}\nabla\phi_i \cdot \nabla\phi_j - \frac{1}{2}\nabla\tilde\phi_i \cdot \nabla\tilde\phi_j\right)$ plus the core parts. Two checks are useful: $\int \tilde\tau\, d^3r$ equals the kinetic energy of the auxiliary wave functions, and $\int(\tau^1 - \tilde\tau^1)\, d^3r = \sum_{ij} D_{ij}\Delta T_{ij}$. The setups of the meta-GGAs are constructed with PBE, as for VASP.

Functional derivatives. For a semilocal energy $E = \int e(n, |\nabla n|^2, \nabla^2 n)\, d^3r$ the potential is the full Euler–Lagrange derivative

$$v = \frac{\partial e}{\partial n} - \nabla \cdot \left(2\frac{\partial e}{\partial |\nabla n|^2}\nabla n\right) + \nabla^2 \frac{\partial e}{\partial \nabla^2 n}. \tag{2.18}$$

For GGAs the last term is absent; for SCAN-L it is essential. Where the energy carries a spatial weight (the double counting of the local hybrids, Sec. 5.6), the weight stands *inside* the derivatives. In the spherical one-center expansion the divergence becomes $r^{-2}(r^2 A)'$; in the non-spherical case the weak form is used, projecting $\partial e/\partial\nabla n$ onto the radial and tangential gradients of each spherical harmonic.

Pitfall. Before September 2026, the one-center SCAN-L and TSCAN potentials kept only the first term of Eq. (2.18). Self-consistent solutions were then not stationary points of the energy: the variational optimizer found lower energies without reaching self-consistency, and for TSCAN it did not converge at all. The consistency test of Sec. 6.7 exposed both.

Pitfall (October 2026). Three errors made all meta-GGA band gaps too small by about 0.3 eV, e.g. 4.93 instead of 5.24 eV for SCAN MgO: the setups were built from an atom without exchange and correlation, the one-center τ lacked the factor $\frac{1}{2}$, and the pseudized core τ was missing in the plane-wave part. The energy and the Hamiltonian were consistent with each other, so the consistency tests passed. They were found with the two checks above and by comparison with VASP: SCAN and r^{2}SCAN gaps now agree within 0.06 eV (MgO 5.30 and 5.34 eV, VASP 5.24 and 5.30 eV; Si 0.949 and 0.919 eV, VASP 0.969 and 0.917 eV). For SCAN0r the double counting also subtracted the potential of the core τ, which made the Hamiltonian inconsistent with the energy and stalled the variational optimizer.

2.7 Spin

Collinear spin polarization (**`!STRUCTURE!OCCUPATIONS NSPIN=2`**) uses two sets of orbitals and the spin densities $n_\uparrow, n_\downarrow$; the code works internally with total and spin density (n, m). Non-collinear magnetism (**`NSPIN=3`**) uses two-component spinors $\psi_n = (\psi_{n\uparrow}, \psi_{n\downarrow})$. The density becomes a 2×2 matrix, which is stored as (n, m_x, m_y, m_z),

$$\rho_{ss'}(\boldsymbol{r}) = \sum_n f_n \psi_{ns}(\boldsymbol{r}) \psi^*_{ns'}(\boldsymbol{r}) = \tfrac{1}{2}\big(n\,\mathbf{1} + \boldsymbol{m}\cdot\boldsymbol{\sigma}\big)_{ss'}, \tag{2.19}$$

with the Pauli matrices $\boldsymbol{\sigma}$, and the Hamiltonian has the form $H = H_0\mathbf{1} + \boldsymbol{B}\cdot\boldsymbol{\sigma}$. The one-center density matrices $D_{R,ij}$ and the matrices $\Delta H_{R,ij}$ of Chapter 4 acquire the same four components. The exchange-correlation energy is evaluated locally with n and $|\boldsymbol{m}|$, and the exchange-correlation field points along the local magnetization. Time reversal does not map a spinor at $\boldsymbol{k}$ onto one at $-\boldsymbol{k}$ in a magnetic system, so the full k-point mesh is used. Spin–orbit coupling (Sec. 4.6) requires this mode. For exchange, the spin-scaling relation

$$E_x[n_\uparrow, n_\downarrow] = \tfrac{1}{2}\big(E_x[2n_\uparrow] + E_x[2n_\downarrow]\big) \tag{2.20}$$

is exact and allows any spin-unpolarized exchange functional to be applied to a spin-polarized density. VESPAW uses it for the exchange of the SCAN family, on the grid and in the one-center terms. Correlation does not obey Eq. (2.20); SCAN correlation is evaluated for the spin densities with the spin interpolation of Ref. [24]: $\phi(\zeta)$ in the gradient term, $G_c(\zeta)$ in the single-orbital limit, and α divided by $d_s(\zeta) = [(1+\zeta)^{5/3} + (1-\zeta)^{5/3}]/2$ with the total τ. For SCAN-L, the spin-resolved kinetic-energy densities are deorbitalized separately, $\tau[n_\uparrow, n_\downarrow] = \frac{1}{2}(\tau[2n_\uparrow] + \tau[2n_\downarrow])$, as in libxc.

2.8 Finite temperature

For metals and for systems with near-degenerate states, VESPAW minimizes the Mermin free energy [29]

$$A = E[\{\psi_n\}, \{f_n\}] - TS[\{f_n\}], \qquad S = -k_B \sum_n \big[f_n \ln f_n + (1-f_n)\ln(1-f_n)\big], \tag{2.21}$$

with Fermi occupations $f_n = [1 + e^{(\epsilon_n - \mu)/k_B T}]^{-1}$ at fixed electron number. The electronic temperature is set in **`!CONTROL!MERMIN T[K]`**. It acts as a smearing parameter; the free energy A, not E, is the variational quantity.

Further reading

The original work on the subject of this chapter is found in Refs. [2, 6–8, 11, 16, 24, 29].

Table 2.1: Exchange-correlation functionals of VESPAW. "libxc" marks functionals whose energy density agrees with libxc 7.1.2 (and the VASP implementation) to $\sim 10^{-15}$ at every point of a grid in density, gradient, spin polarization and α (October 2026). The local hybrids are described in Chapter 5.

TYPE	name	functional	VASP	libxc
LDA				
1	**PZ**	Perdew–Zunger [13] of Ceperley–Alder [12]		✓
2	**PW92**	Perdew–Wang 92 [14]		✓
3		Slater exchange only		✓
4		X_α, $\alpha = 0.7$		
GGA				
10	**PBE**, **PE**	PBE [16]	**GGA=PE**	✓
11	**RPBE**, **RP**	RPBE [17]	**GGA=RP**	✓
12	**PW91**, **91**	PW91 exchange and correlation [18]	**GGA=91**	✓
13	**REVPBE**, **RE**	revPBE, $\kappa = 1.245$ [19]	**GGA=RE**	✓
14	**PBESOL**, **PS**	PBEsol [20]	**GGA=PS**	✓
5001	**BLYP**	B88 [21] + LYP [22]		✓
81		BP86: PZ + B88 + P86 [23]		✓
8		B88 + P86 on PW92 local correlation (non-standard)		
7, 71, 9		B88 exchange with PW92, PZ or PBE correlation		✓
6		B88 exchange only		✓
meta-GGA				
21	**SCAN**	SCAN, orbital τ [24] (with **TSCANDELOC=T**)	**METAGGA=SCAN**	✓
22	**R2SCAN**	r^{2}SCAN, orbital τ [25] (with **TSCANDELOC=T**)	**METAGGA=R2SCAN**	✓
20	**SCANL**	SCAN-L, deorbitalized [26, 27]	**METAGGA=SCANL**	✓
local hybrids (Chapter 5)				
	PBE0r	PBE + onsite exact exchange		
	PBE-SL0r	PBE; onsite SCAN-L/exact exchange		
	SCAN0r	SCAN + onsite exact exchange		
	r^{2}SCAN0r	r^{2}SCAN + onsite exact exchange		
for testing and reproducing old results				
−10		PBE before the 2024/2026 corrections (see text)		
−12		PBE exchange + PW91 correlation (old type 12)		
5002		PBE with exchange scaled by 0.75		
10001–7		single components (PZ, PW92, P86, PBE, PW91 correlation; PBE, RPBE exchange)		

Chapter 3

Plane waves and Brillouin-zone sampling

3.1 Introduction 11
3.2 Bloch states and plane waves 11
3.3 Cutoffs and FFT grids 12
3.4 k-point sampling 12
3.5 Space-group symmetry 12
3.6 The Γ point 13
3.7 Slabs, molecules and the dipole correction 13

3.1 Introduction

Plane waves are the simplest basis set one can imagine. They are orthonormal, they do not depend on the positions of the atoms, and their completeness is controlled by a single number, the cutoff. The fast Fourier transform switches between real and reciprocal space at low cost, so that each term of the Hamiltonian can be applied where it is simplest. The price is that sharp features, such as the oscillations of the wave functions near a nucleus, would require an enormous number of plane waves. The PAW method of the next chapter removes this problem. Here we collect what the plane-wave description of a periodic solid involves: Bloch states, cutoffs, the grids for the density, and the sampling of the Brillouin zone.

3.2 Bloch states and plane waves

In a crystal with lattice vectors $\boldsymbol{T}$ the orbitals are Bloch states, $\psi_{n\boldsymbol{k}}(\boldsymbol{r}+\boldsymbol{T}) = e^{i\boldsymbol{k}\cdot\boldsymbol{T}}\psi_{n\boldsymbol{k}}(\boldsymbol{r})$, which are expanded in plane waves,

$$\tilde{\psi}_{n\boldsymbol{k}}(\boldsymbol{r}) = \sum_{\boldsymbol{G}} c_{n\boldsymbol{k}}(\boldsymbol{G})\, e^{i(\boldsymbol{k}+\boldsymbol{G})\cdot\boldsymbol{r}}, \qquad \tfrac{1}{2}|\boldsymbol{k}+\boldsymbol{G}|^2 \le E_{\text{cut}}. \tag{3.1}$$

In PAW only the smooth pseudo wave functions $\tilde{\psi}$ are expanded in plane waves (Chapter 4).

3.3 Cutoffs and FFT grids

The wave-function cutoff is **!CONTROL!FOURIER EPWPSI** in Rydberg. The density contains products of wave functions and would need four times the cutoff; VESPAW uses **EPWRHO = CDUAL**×**EPWPSI**. **CDUAL=4** is exact, **CDUAL=2** is usually sufficient and cheaper. Density and local potential live on a real-space grid; the Hamiltonian is applied to a wave function by transforming it to real space with a fast Fourier transform, multiplying by the potential and transforming back. In VESPAW these transforms are batched over blocks of 32 bands.

Energies converge from above with the cutoff, but energy *differences*, forces and stress converge faster. The required cutoff is set by the hardest PAW setup in the system; the setup generator prints an estimate (Sec. 4.5).

3.4 k-point sampling

Brillouin-zone integrals are replaced by sums over a Monkhorst–Pack mesh [30], **!STRUCTURE!KPOINTS DIV=**n_1 n_2 n_3, with an optional shift, or by a target density **R**. Time reversal ($\psi_{-\boldsymbol{k}} = \psi_{\boldsymbol{k}}^*$) halves the mesh for collinear spin; the space group reduces it further (Sec. 3.5). Metals need dense meshes together with a finite electronic temperature; insulators converge quickly.

3.5 Space-group symmetry

With **!STRUCTURE!KPOINTS SYMMETRY=T** the code uses the space group of the structure. An operation $\{\mathsf{M}|\boldsymbol{t}\}$ maps $\boldsymbol{r}$ onto $\mathsf{M}\boldsymbol{r} + \boldsymbol{t}$. The operations are found from the lattice and the atoms: M must leave the metric of the lattice invariant, and every atom must be mapped onto an atom of the same species. Atoms of one species that carry different orbital-potential seeds (**!ORBPOT**, e.g. the two Ni sublattices of antiferromagnetic NiO) count as different, so that the symmetry of the magnetic order, not that of the lattice, is used.

The mesh is then reduced to its irreducible wedge. Each irreducible point $\boldsymbol{k}$ carries the weight of its star $\{\mathsf{M}\boldsymbol{k}\}$. Quantities built from a sum over the irreducible points have the full symmetry only after symmetrization, which follows the scheme of VASP:

- the pseudo density, $\tilde{n}(\boldsymbol{r}) = \frac{1}{N_{\mathrm{op}}} \sum_g \tilde{n}_{\mathrm{IBZ}}(\mathsf{M}_g\boldsymbol{r} + \boldsymbol{t}_g)$, exactly on the real-space grid when every operation maps grid points onto grid points, otherwise in reciprocal space, where the coefficient at $\mathsf{M}^{\mathsf{T}}\boldsymbol{G}$ is $\tilde{n}(\boldsymbol{G})e^{i\boldsymbol{G}\cdot\boldsymbol{t}}$;
- the one-center density matrices, $D_{R'} = \frac{1}{N_{\mathrm{op}}} \sum_g U_g D_R U_g^{\mathsf{T}}$ with R' the image of atom R and U_g the rotation matrix of the real spherical harmonics (times $(-1)^\ell$ for improper operations);
- forces, which transform as vectors, and the stress, which transforms as a tensor.

A symmetric run must reproduce the full-mesh run; the regression cases `symmetry_al` (48 operations, grid symmetrized in real space) and `symmetry_si` (displaced Si, 8 operations with non-lattice translations, symmetrized in reciprocal space) check energies, forces and stress. Symmetry is not available with non-collinear spin, onsite hybrids or delocalized SCAN.

Pitfalls. Two bugs that the symmetry code uncovered also affected ordinary runs. At time-reversal-invariant k-points, two real bands are packed into one complex array (Sec. 3.6); exactly degenerate bands came out of the subspace diagonalization as complex mixtures and were destroyed by the packing, which led to energies below the minimum. The subspace matrices are now made real at these points. The strain derivative of the projectors was not symmetric, so the off-diagonal stress was wrong; this was found by comparing per-k contributions across the

members of a star, which must transform into each other. A dense metallic mesh with symmetry (fcc Al, 12^3) still needs several times more line-search iterations than the full mesh, although it converges to the same energy; the cause is open.

3.6 The Γ point

At $\boldsymbol{k} = 0$ the wave functions can be chosen real. CP-PAW exploits this by packing two real bands into one complex array ("super wave functions"), which halves the number of FFTs. An alternative half-sphere storage (**`GAMMASUPER`**) is experimental and not yet correct.

3.7 Slabs, molecules and the dipole correction

Molecules and surfaces are treated in a periodic cell with vacuum between the images. A slab with different top and bottom surfaces, or with an adsorbate on one side, carries a dipole moment per area. The periodic boundary conditions force the potential to be periodic, so the images create an artificial uniform field across the vacuum and across the slab, which shifts work functions and adsorption energies and converges only slowly with the thickness of the vacuum. The dipole correction of Neugebauer and Scheffler [31], in the form of Bengtsson [32], removes this field (**`!CONTROL!DFT DIPOLECORR=T`**).

The slab normal is the third lattice vector, $L_3 = V/\mathcal{S}$ is the height of the cell, V its volume and $\mathcal{S}$ the area of its base. A sawtooth function $z(s) = L_3\big(\{s - s_0\} - \tfrac{1}{2}\big)$, where s is the fractional coordinate along the normal and $\{\cdot\}$ the fractional part, is continuous everywhere except for a step at s_0, which is placed in the middle of the largest gap between atomic planes. With the dipole moment of the total charge density ρ (electrons, nuclei and compensation charges)

$$m = \int_V d^3r\, \rho(\boldsymbol{r})\, z(\boldsymbol{r}), \tag{3.2}$$

the correction adds

$$E_{\mathrm{dip}} = \frac{2\pi m^2}{V}, \qquad v_{\mathrm{dip}}(\boldsymbol{r}) = \frac{4\pi m}{V}\, z(\boldsymbol{r}) \tag{3.3}$$

to the electrostatic energy and potential. The linear potential cancels the artificial field, and the potential jumps by $4\pi m/\mathcal{S}$ at the step (Fig. 3.1). The sawtooth has nonzero Fourier coefficients only on the line $\boldsymbol{G} = h\boldsymbol{b}_3$, $z(h) = iL_3/(2\pi h)\, e^{-2\pi i h s_0}$, so both m and v_{dip} are computed in reciprocal space at negligible cost. The potential step is printed as `DIPOLE POTENTIAL STEP`; it is the difference of the work functions of the two surfaces. Forces include the correction, because it acts on the compensation charges through the electrostatic potential. The stress is not corrected.

The regression case `h2o_dipole` is a polar water layer in an 18 Å cell. E_{dip} is 1.36 mH at 12 Å and 0.89 mH at 18 Å, and scales with $1/V$ as it should. The same case guards the variational optimizer in a large vacuum (Sec. 6.4), where $\boldsymbol{\theta}$ is set to the output potential at points without density and the mean potential can move by hartrees during the iteration.

Further reading

The original work on the subject of this chapter is found in Refs. [30–33].

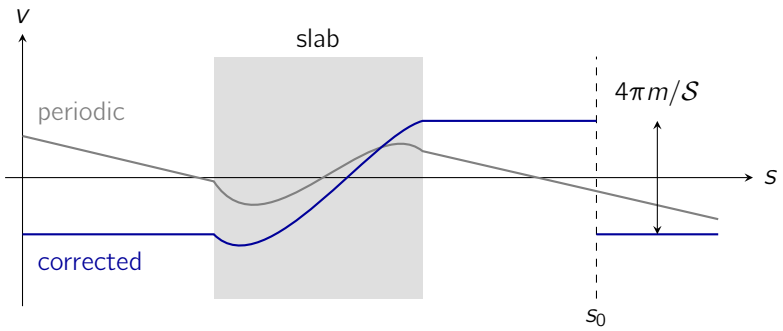


Figure 3.1: Planar average of the electrostatic potential across a polar slab (schematic). Without the correction (gray) the periodic images create a uniform field in the vacuum. The sawtooth potential of Eq. (3.3) makes the potential flat in the vacuum on both sides and moves the jump to the step at s_0 (blue).

Chapter 4

The projector augmented-wave method

4.1 Introduction 15
4.2 The PAW transformation 15
4.3 Compensation charges, core and local potential 17
4.4 Setup construction 17
4.5 Quality checks 17
4.6 Relativistic effects 18
4.7 The setup library 19
4.8 Core-level spectra 19

4.1 Introduction

A wave function in a solid has two faces. Between the atoms, where the chemical bonds form, it is smooth. Close to a nucleus it oscillates rapidly, because it must be orthogonal to the core states. The classic electronic-structure methods have chosen one face. Pseudopotentials replace the atomic region by a smooth effective potential and lose the true wave function there. Augmented-wave methods keep it, but at the price of a basis set that depends on energy and on the atomic positions. The PAW method of Blöchl unites both views through a linear transformation between smooth auxiliary wave functions, which are expanded in plane waves, and the true wave functions. This chapter describes the transformation, the construction of the PAW setups, and the checks that make a setup trustworthy.

4.2 The PAW transformation

The PAW method [1] generalizes the ideas of the pseudopotential [34] and of the linear augmented-wave methods [35]. It maps the true all-electron (AE) valence orbitals ψ_n, which oscillate strongly near the nuclei, onto smooth pseudo (PS) orbitals $\tilde{\psi}_n$ by a linear transformation $|\psi_n\rangle = \mathcal{T}|\tilde{\psi}_n\rangle$. The transformation differs from unity only inside augmentation spheres around the atoms, $\mathcal{T} = 1 + \sum_R \mathcal{T}_R$. In each sphere we choose AE partial waves $\phi_{R,i}$ (solutions of the atomic Schrödinger equation at reference energies) and PS partial waves $\tilde{\phi}_{R,i}$, which agree with $\phi_{R,i}$ outside an augmentation radius r_c, such that $|\phi_{R,i}\rangle = \mathcal{T}|\tilde{\phi}_{R,i}\rangle$. Inside the sphere a PS orbital is expanded in PS partial waves, $|\tilde{\psi}_n\rangle = \sum_i |\tilde{\phi}_{R,i}\rangle c_{n,i}$. Because $\mathcal{T}$ is linear, the AE orbital has the same coefficients in the AE partial waves, $\mathcal{T}|\tilde{\psi}_n\rangle = \sum_i |\phi_{R,i}\rangle c_{n,i}$. The coefficients are obtained with

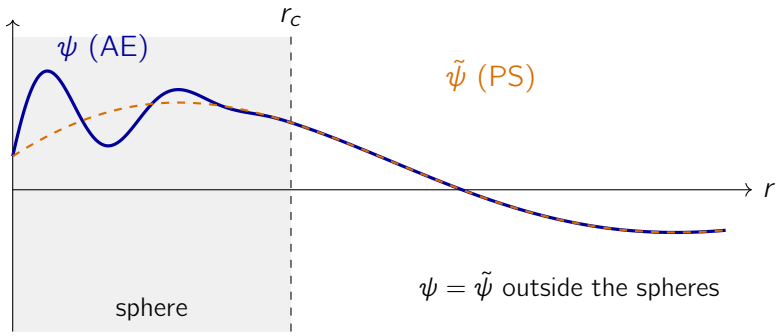


Figure 4.1: The PAW transformation, Eq. (4.1) (schematic). Outside the augmentation sphere the true wave function ψ (solid) and the smooth pseudo wave function $\tilde{\psi}$ (dashed) coincide. Inside, ψ oscillates, because it is orthogonal to the core states, while $\tilde{\psi}$ stays smooth and can be expanded in few plane waves. The difference is carried by the partial waves on radial grids.

projector functions $\tilde{p}_{R,i}$, localized within r_c and dual to the PS partial waves, $\langle \tilde{p}_{R,i} | \tilde{\phi}_{R,j} \rangle = \delta_{ij}$, as $c_{n,i} = \langle \tilde{p}_{R,i} | \tilde{\psi}_n \rangle$. This gives

$$\mathcal{T} = 1 + \sum_{R,i} \big(|\phi_{R,i}\rangle - |\tilde{\phi}_{R,i}\rangle \big) \langle \tilde{p}_{R,i}|, \qquad |\psi_n\rangle = |\tilde{\psi}_n\rangle + \sum_{R,i} \big(|\phi_{R,i}\rangle - |\tilde{\phi}_{R,i}\rangle \big) \langle \tilde{p}_{R,i} | \tilde{\psi}_n \rangle. \tag{4.1}$$

In words: inside each sphere the smooth part $\sum_i |\tilde{\phi}_{R,i}\rangle \langle \tilde{p}_{R,i} | \tilde{\psi}_n \rangle$ is removed and replaced by its AE counterpart (Fig. 4.1). The expansion is exact for complete sets of partial waves; in practice one or two partial waves per angular momentum suffice. The index i collects the angular momentum (ℓ, m) and a counter ν over partial waves of the same ℓ. The number of partial waves per ℓ is set by **`!STRUCTURE!SPECIES NPRO`**.

Frozen core. The core states are taken from the isolated atom and kept frozen, $|\psi^c_{R,\alpha}\rangle = |\phi^c_{R,\alpha}\rangle$, where α runs over the core states of atom R. They are not strictly localized within the sphere. Expectation values of a local operator A then split into a plane-wave part, one-center corrections and a core term,

$$\langle A \rangle = \sum_n f_n \langle \tilde{\psi}_n | \tilde{A} | \tilde{\psi}_n \rangle + \sum_R \sum_{ij} D_{R,ij} \big(\langle \phi_{R,j} | A | \phi_{R,i} \rangle - \langle \tilde{\phi}_{R,j} | \tilde{A} | \tilde{\phi}_{R,i} \rangle \big) + \sum_{R,\alpha} \langle \phi^c_{R,\alpha} | A | \phi^c_{R,\alpha} \rangle,$$
$$D_{R,ij} = \sum_n f_n \langle \tilde{p}_{R,i} | \tilde{\psi}_n \rangle \langle \tilde{\psi}_n | \tilde{p}_{R,j} \rangle, \tag{4.2}$$

with the one-center density matrix $D_{R,ij}$. For the density ($A = |\boldsymbol{r}\rangle\langle\boldsymbol{r}|$) this gives

$$n(\boldsymbol{r}) = \tilde{n}(\boldsymbol{r}) + \sum_R \big(n^1_R(\boldsymbol{r}) - \tilde{n}^1_R(\boldsymbol{r}) \big), \qquad n^1_R = \sum_{ij} D_{R,ij}\, \phi_{R,j} \phi_{R,i} + n^c_R, \tag{4.3}$$

where $\tilde{n} = \sum_n f_n |\tilde{\psi}_n|^2$ (plus the pseudized core density) is smooth and represented on the plane-wave grid, while the one-center densities n^1_R and $\tilde{n}^1_R$ are represented on radial grids times spherical harmonics. The total energy becomes $E = \tilde{E} + \sum_R (E^1_R - \tilde{E}^1_R)$: a plane-wave term plus, for each atom, the difference of an AE and a PS one-center energy evaluated on radial grids. The orbitals are orthonormal with respect to the overlap operator

$$\hat{O} = 1 + \sum_{R,ij} |\tilde{p}_{R,i}\rangle \Delta O_{R,ij} \langle \tilde{p}_{R,j}|, \qquad \Delta O_{R,ij} = \langle \phi_{R,i} | \phi_{R,j} \rangle - \langle \tilde{\phi}_{R,i} | \tilde{\phi}_{R,j} \rangle, \tag{4.4}$$

so the Kohn–Sham equations take the generalized form $\tilde{H}|\tilde{\psi}_n\rangle = \epsilon_n \hat{O} |\tilde{\psi}_n\rangle$ with

$$\tilde{H} = -\tfrac{1}{2}\nabla^2 + \tilde{v}(\boldsymbol{r}) + \sum_{R,ij} |\tilde{p}_{R,i}\rangle \Delta H_{R,ij} \langle \tilde{p}_{R,j}|. \tag{4.5}$$

4.3 Compensation charges, core and local potential

The electrostatics of the one-center densities is handled by compensation charges $\hat{n}_R$ (Gaussians of width set by **RCSM/RCOV**) that reproduce the multipole moments of $n^1_R - \tilde{n}^1_R$, so that the long-range electrostatics is carried by the smooth density $\tilde{n} + \hat{n}$ on the plane-wave grid. The one-center expansions are truncated at $\ell \leq$ **LRHOX**. Core electrons are frozen in their atomic state. For the exchange-correlation energy, which is non-linear in the density, a pseudized core density $\tilde{n}^c_R$ (**!CORE POW RC**) is added to the smooth density. The local pseudopotential $\bar{v}_R$ is constructed by pseudizing the AE potential inside a radius (**!POT POW RC**).

4.4 Setup construction

A PAW *setup* contains, for one element, the partial waves, projectors, core densities and local potential. VESPAW builds setups at run time from a short parameter block (**!SPECIES!AUGMENT**), or reads them from the library `vespaw_pot`. Three construction types exist:

- **HBS**: the scheme of Blöchl's later CP-PAW versions. The first partial wave of each ℓ is the lowest valence state; further partial waves are obtained from the nodeless ("$\dot{\phi}$-like") construction at higher energies, bounded by a box of radius **RBOX/RCOV**.
- **NDLSS**: an older nodeless scheme.
- **KERKER**: Kerker-type pseudization.

The radii are given in units of the covalent radius of the element: **RCL/RCOV** for the partial waves, **RCSM/RCOV** for the compensation charge, **RC/RCOV** in **!POT** and **!CORE**.

Semicore states. When shallow core states are treated as valence (e.g. Cu 3*s*3*p*, **ZV=19**), an angular momentum has two valence states, 3*s* and 4*s*. Each valence bound state then receives its own partial wave at its own energy; only further partial waves are box states. With a semicore state below it, the PS 4*s* orbital need not have a node, because it is orthogonal to the PS 3*s* only through $\hat{O}$. The setup test therefore identifies PS states by their energy, not by their node count.

4.5 Quality checks

A setup is an approximation, and a bad one fails silently: calculations converge to wrong answers. The generator therefore reports three checks.

Ghost test. The atomic PAW Hamiltonian of each ℓ is diagonalized in a basis of 120 spherical Bessel functions inside a large sphere. Its lowest levels must match the AE valence levels. A level far below the lowest AE reference is a *ghost state*; the generator prints `WARNING: GHOST STATE IN SETUP`. The overlap matrix in this basis must be positive definite; if it is not, the setup is unusable.

Incompleteness warning. With only one partial wave per ℓ, the local potential may bind a state far below the valence level whose wave function lies largely outside the span of the projectors. Such components receive no one-center correction. In a solid they can sink into the potential well of the compensation charge, and the calculation converges to a collapsed state far below the physical minimum. This happened for NiO with a single-projector Ni setup: the collapse deepened with the cutoff and disappeared with a second *s* projector. The generator prints `WARNING: INCOMPLETE SETUP` when the lowest state of the local potential lies more than 0.5 H below the valence reference with a single projector.

Kinetic-energy cutoff table. As in the VASP setup generator, the kinetic energy of each PS valence orbital that lies above a cutoff is computed from its Fourier transform, weighted by the occupation and summed. The generator prints the error as a function of the cutoff, from 20 eV to about 290 Ry, and suggests `ENMAX` and `ENMIN` (errors of 6 and 13 meV), or states that they are not reached. The PS orbitals are tapered to zero before the box wall to avoid a spurious kink.

4.6 Relativistic effects

Relativistic effects are large only close to the nucleus, where the kinetic energy of the electrons is large. In PAW this region lies inside the augmentation spheres, so relativity enters through the partial waves and the one-center terms. The plane-wave part stays non-relativistic.

Scalar-relativistic partial waves. Dropping spin–orbit coupling from the Dirac equation leaves the scalar-relativistic equation of Koelling and Harmon [36] for the large component,

$$\Big[-\tfrac{1}{2}\nabla\cdot\frac{1}{M(r)}\nabla + v(r)\Big]\phi = \epsilon\phi, \qquad M(r) = 1 + \frac{\epsilon - v(r)}{2c^2}, \tag{4.6}$$

which contains the mass-velocity and Darwin terms; the code stores $d_{\mathrm{rel}} = 1/M - 1$. Because M depends on the energy, every state of the atom has its own operator. This is fine for the self-consistent atom, which is solved fully scalar-relativistically. For the partial waves it is not: partial waves of one ℓ at different energies would see different kinetic-energy operators, and the one-center kinetic-energy matrix $\langle\phi_i|\hat{T}|\phi_j\rangle$ would not be symmetric. VESPAW therefore uses one energy-independent M per angular momentum, evaluated at E_ν, the highest valence level of that ℓ. The valence levels of the atom are then reproduced to 2×10^{-4} eV, deep semicore levels to tens of meV (Pt 5s: 53 meV, Rh 4s: 30 meV). The generator prints the comparison for every level (`SETUP LEVEL CHECK`). Symmetrizing the energy-dependent matrix instead had placed the Pt 5s level 11.7 eV off, and the Pt and Rh setups could not be built. The zeroth-order regular approximation (ZORA [37], **`!AUGMENT ZORA=T`**) sets $E_\nu = 0$; it contracts the lattice constant of Cu by 0.4%. The default is the scalar-relativistic treatment, as in VASP.

Spin–orbit coupling. The spin–orbit term that was dropped from Eq. (4.6) is, for a spherical potential,

$$\hat{H}_{\mathrm{SO}} = \xi(r)\,\boldsymbol{L}\cdot\boldsymbol{S}, \qquad \xi(r) = \frac{1}{2M^2c^2}\,\frac{1}{r}\frac{dv}{dr}, \qquad M = 1 - \frac{v}{2c^2}. \tag{4.7}$$

It is largest near the nucleus, so it is evaluated in the spheres only, between AE partial waves and with the spherical AE potential of the isolated atom. It adds a spinor term to the one-center Hamiltonian,

$$\Delta H^{\mathrm{SO}}_{R,(\nu\ell ms),(\nu'\ell m's')} = \Xi_{R,\nu\nu'}\,\langle\ell m|\boldsymbol{L}|\ell m'\rangle\cdot\tfrac{1}{2}\boldsymbol{\sigma}_{ss'}, \qquad \Xi_{R,\nu\nu'} = \int_0^{r_c} dr\, r^2\,\phi_{R,\nu}(r)\,\xi(r)\,\phi_{R,\nu'}(r), \tag{4.8}$$

with r_c the covalent radius and $\ell \le 3$. The energy is $\mathrm{Tr}[D\,\Delta H^{\mathrm{SO}}]$, so the term enters the self-consistent spinor calculation like any other part of ΔH; there is no second variation. Spin–orbit coupling requires non-collinear spin (**`!CONTROL!DFT SPINORBIT=T`** with **`NSPIN=3`**). The partial waves are scalar-relativistic, so the $p_{1/2}$ and $p_{3/2}$ radial functions are the same, as in VASP. For Si the valence band at Γ splits by $\Delta_{\mathrm{so}} = 48$ meV (experiment 44 meV; regression case `si_soc`).

Pitfall. The spin-flip block of the spinor Hamiltonian in CP-PAW was built as $\Delta H_{\downarrow\uparrow} = (\Delta H_{\uparrow\downarrow})^*$, which is correct only when ΔH_x and ΔH_y are real. Spin–orbit coupling makes them complex; the block was then anti-Hermitian and the coupling silently absent. It is now built as $\Delta H_x + i\Delta H_y$.

4.7 The setup library

`vespaw_pot` contains one species block per element and variant. Its audit (`vespaw_pot/AUDIT.md`) lists each setup with its status (OK, incomplete, ghost, unusable) and the recommended replacement. The flagged setups were rebuilt as HBS setups with two projectors per ℓ. Validation in solids shows that the frozen-semicore $3d$ setups (Fe, Cu) reproduce AE-PBE lattice constants only to about 1%, and that the error grows with the augmentation radius; a semicore Cu setup with small radii reaches 0.5% at a high cutoff. Accurate soft $3d$ setups are an open problem (see the audit).

4.8 Core-level spectra

Because PAW keeps the AE partial waves and the frozen core states, matrix elements between core and band states are available. VESPAW uses them for electron-energy-loss near-edge spectra (ELNES), with **`!CONTROL!ANALYSE!EELS`**. The same analysis applies to x-ray absorption near-edge spectra (XANES), which also probe the excitation of a core electron into empty states. The spectrum is the loss function

$$L(\omega) = -\mathrm{Im}\left[\epsilon_r^{-1}(\omega)\right], \qquad \epsilon_r(\omega) = 1 + 4\pi\frac{N}{V}\sum_n (1-f_n)\frac{Q_n}{(\epsilon_n-\epsilon_c)^2-\omega^2-i\Gamma\omega}, \tag{4.9}$$

following Wooten [38], where N/V is the number density of the probed atom, ϵ_c the core level and ϵ_n the energy of an empty Kohn–Sham state. The oscillator strengths are

$$Q_n = 2\sum_c (\epsilon_n-\epsilon_c)\,\frac{1}{3}\sum_{j=1}^{3}\left|\langle\psi_c|x_j|\psi_n\rangle\right|^2, \tag{4.10}$$

where c runs over the states of the core shell (**`IC`**). The core state is localized, so the band state can be replaced by its one-center expansion, $|\psi_n\rangle \approx \sum_i |\phi_{R,i}\rangle\langle\tilde p_{R,i}|\tilde\psi_n\rangle$. With $\phi_{R,i}(\boldsymbol{r}) = R_i(r)Y_{L_i}(\hat r)$ and the dipole operator written with real spherical harmonics, the matrix elements reduce to radial integrals and Gaunt coefficients $C_{L,L',L''} = \int d\Omega\, Y_L Y_{L'} Y_{L''}$,

$$\langle\psi_c|x_j|\psi_n\rangle = \sqrt{\frac{4\pi}{3}}\sum_i C_{L_c,p_j,L_i}\int dr\, r^3 R_c(r)R_i(r)\,\langle\tilde p_{R,i}|\tilde\psi_n\rangle, \tag{4.11}$$

where p_j denotes the $\ell = 1$ harmonic along x_j. The spectrum therefore measures the empty density of states on the probed atom with angular momentum $\ell_c \pm 1$.

Broadening. The width Γ is the sum of the core-hole and the band-state lifetime broadenings. The core-hole width of a K shell is taken from Eq. 12 of Egerton [39], $\Gamma_K = [-0.285 + 0.0216\,(E_{\mathrm{thr}}/\mathrm{eV})^{0.472}]$ eV, with the threshold energy $E_{\mathrm{thr}} = -\epsilon_c$; other shells use Eq. 14 of the same reference. The band state with energy ϵ above the Fermi level decays with the rate v/λ, where $v = \sqrt{2\epsilon}$ is the free-electron velocity and λ the inelastic mean free path, interpolated as $\lambda = A/\epsilon^2 + B\sqrt{\epsilon}$ with $A = 641$ eV^2 nm and $B = 0.096$ nm eV$^{-1/2}$ for inorganic compounds [40]. The result is finally convoluted with a Gaussian for the instrumental resolution (**`BROADENING[EV]`**).

Limitations. The calculation uses the independent-particle approximation and ignores the core hole; its only effect is a rigid shift of the spectrum, which must be adjusted to the experimental threshold. A better approximation is a supercell calculation with half a core hole on the probed atom (Slater's transition-state rule), and a still better one is the solution of the Bethe–Salpeter

equation [41]. The independent-particle approximation is adequate for the oxygen K edge ($1s \rightarrow 2p$), where the core hole is strongly screened [42]. It is not adequate for transition-metal $L_{2,3}$ edges, whose shape is dominated by multiplet effects from the overlap of the $2p$ hole with the $3d$ electrons.

Further reading

The original work on the subject of this chapter is found in Refs. [1, 36, 39, 43].

Chapter 5

Local hybrid functionals

5.1 Introduction . . . 21
5.2 Hybrid functionals . . . 21
5.3 Local orbitals . . . 22
5.4 Projection onto the local orbitals . . . 22
5.5 PBE0r . . . 22
5.6 PBE-SL0r . . . 23
5.7 SCAN0r . . . 23
5.8 DFT+U . . . 24

5.1 Introduction

Hybrid functionals admix a fraction a_x of exact exchange and cure much of the band-gap problem of semilocal functionals, but in a plane-wave basis exact exchange is one to two orders of magnitude more expensive than a GGA. Local hybrids of the PBE0r type [44, 45] keep only the onsite part of the exchange correction, evaluated in a basis of local orbitals, at nearly semilocal cost.

5.2 Hybrid functionals

The adiabatic connection, Eq. (2.13), writes the exchange-correlation energy as an integral of $U_{\mathrm{xc}}^{\lambda}$ from the non-interacting ($\lambda = 0$) to the fully interacting ($\lambda = 1$) system. At $\lambda = 0$ the integrand is the exact exchange energy of the Kohn–Sham determinant; near $\lambda = 1$ a semilocal functional is a good approximation. Becke [46] approximated the integral by the trapezoidal rule, $E_{\mathrm{xc}} \approx \frac{1}{2}E_x^{\mathrm{HF}} + \frac{1}{2}U_{\mathrm{xc}}^{\lambda=1}$, and obtained the "half-and-half" functional. If the integrand is a convex rather than a straight function of λ, a smaller weight of exact exchange is appropriate. The general form of a hybrid is

$$E_{\mathrm{xc}}^{\mathrm{hybrid}} = E_{\mathrm{xc}}^{\mathrm{GGA}} + a_x\left(E_x^{\mathrm{HF}} - E_x^{\mathrm{GGA}}\right). \tag{5.1}$$

Perdew, Ernzerhof and Burke [47] argued from perturbation theory that $a_x = \frac{1}{4}$ is a good choice with PBE, which gives PBE0 [48]. The most widely used hybrid in chemistry, B3LYP [49], is a generalization of Becke's three-parameter form [50]. Hybrids improve binding energies, reaction energies and band gaps over GGAs. In transition-metal oxides they open the gaps that semilocal functionals close, because exact exchange stabilizes occupied orbitals relative to empty orbitals of the same character, and removes the self-interaction of the localized d and f electrons.

5.3 Local orbitals

Each atom carries natural tight-binding orbitals (NTBOs) χ_a [44, 51]. Inside the augmentation sphere of radius **RAUG/RCOV** the head function of an orbital is an augmented screened Hankel function, a combination of an AE partial wave ϕ and its energy derivative $\dot\phi$,

$$K^{\mathrm{aug}} = \phi\, k_\phi + \dot\phi\, k_{\dot\phi}, \tag{5.2}$$

matched to a Hankel function with κ^2 set by **K2**. In the spheres of the neighbors it is continued by screened Bessel functions $\bar J^{\mathrm{aug}} = \dot\phi\, j_{\dot\phi}$ with the structure constants $\bar S$. Exponential tails (**TAILLAMBDA**, **RTAIL/RCOV**) and a cutoff (**RTAILCUT/RCOV**) make the orbitals short ranged. The orbitals considered per ℓ are selected by **NOFL**.

The construction follows the tight-binding linear muffin-tin orbital method [35, 51]: only the nodeless scattering partial waves are admitted on the neighboring sites, the same framework of nodeless partial waves that is used for the augmentation [52].

5.4 Projection onto the local orbitals

The onsite density matrix $D^\sigma_{ab} = \sum_n f_{n\sigma}\langle\chi_a|\psi_{n\sigma}\rangle\langle\psi_{n\sigma}|\chi_b\rangle$ requires the coefficients of the wave functions in the local-orbital basis. Inside the spheres the one-center expansion $\psi^1 = \sum_i \phi_i\langle\tilde p_i|\tilde\psi\rangle$ is fitted by local orbitals in the least-squares sense,

$$\boldsymbol{c} = \big[\langle\chi|\Theta|\chi\rangle\big]^{-1}\langle\chi|\Theta|\phi\rangle\langle\tilde p|\tilde\psi\rangle, \tag{5.3}$$

where Θ restricts integrals to the spheres. Both overlaps are computed with the exact χ inside the spheres, i.e. with K^{aug} and $\bar J^{\mathrm{aug}}$ as radial functions.

Pitfall. The original implementation represented χ by its partial-wave expansion $\langle\tilde p|\tilde\chi\rangle$. The $\dot\phi$ component of χ converges slowly in that expansion, so the fitted orbital lacked norm and the density matrix overcounted charge: 13–15 electrons on Ni in NiO instead of about 7. The hybrid energy and the gap then changed with the number of projectors (NiO PBE0r gap 3.96, 4.08, 4.10 eV for 2, 3, 4 d projectors). With the exact overlaps they are converged (2.99, 3.00, 3.00 eV). The old fit can be restored with the environment variable `VESPAW_NTBO_OLDPROJ=1` for comparison.

5.5 PBE0r

PBE0r keeps only the onsite terms of the PBE0 exchange correction,

$$E_{\mathrm{xc}}^{PBE0r} = E_{\mathrm{xc}}^{\mathrm{PBE}}[n] + a_x\big(E_x^{\mathrm{HF,1c}} - E_x^{\mathrm{PBE,1c}}\big), \qquad E_x^{\mathrm{HF,1c}} = -\frac{1}{2}\sum_\sigma\sum_{abcd} D^\sigma_{ac} D^\sigma_{bd} U_{abcd}, \tag{5.4}$$

with the onsite Coulomb tensor U of the local orbitals. The double-counting term $E_x^{\mathrm{PBE,1c}}$ is the PBE exchange of the local-orbital density, obtained from the full one-center density with the weight $w(r) = (n_{\mathrm{loc}}/n_{\mathrm{all}})^2$. Core–valence exchange can be included (**CV=T**). The mixing a_x is set globally by **!CONTROL!DFT!NTBO HFWEIGHT** and per species by **LHFWEIGHT**. The exchange correction enters the Hamiltonian in two places: the one-center PAW Hamiltonian ΔH (derivative with respect to the full one-center density) and an operator in the local-orbital basis (derivative with respect to D).

Design choices. In contrast to other local hybrids [53, 54], which correct only the d or f shell of selected atoms, PBE0r includes the correction on all atoms and for all local orbitals that contribute to the valence or contain core states, including the exchange between core and valence electrons. Applying the correction to a subset of orbitals would shift them in energy relative to the uncorrected ones; including all orbitals avoids such artificial shifts. Because the correction is onsite, the mixing factor can be chosen per atom, which reflects an effective local dielectric screening. The common value $a_x = 0.25$ is not always appropriate: in Pr manganites it made the observed configuration of the Pr $4f$ shell unstable, and the mixing was adjusted per species to reproduce the observed spectral features [45].

5.6 PBE-SL0r

PBE-SL0r (called SCAN0r before October 2026) replaces the semilocal onsite exchange by SCAN-L exchange while keeping PBE for the delocalized part and for all correlation:

$$\begin{aligned} E_{\mathrm{xc}}^{PBE-SL0r} &= E_{\mathrm{xc}}^{\mathrm{PBE}}[n] - E_x^{\mathrm{PBE,1c}} + a_x E_x^{\mathrm{HF,1c}} + (1-a_x) E_x^{\mathrm{SCAN,1c}} \\ &= E_{\mathrm{xc}}^{\mathrm{PBE}}[n] + \underbrace{\left(E_x^{\mathrm{SCAN,1c}} - E_x^{\mathrm{PBE,1c}}\right)}_{\text{swap term}} + a_x\left(E_x^{\mathrm{HF,1c}} - E_x^{\mathrm{SCAN,1c}}\right). \end{aligned} \tag{5.5}$$

The swap term is not scaled by a_x. PBE-SL0r is selected by **`!NTBO MODUS='HYBRID' HFWEIGHT=`**a_x **`DCTYPE='SCANL'`** with a PBE base functional. A consistent implementation requires:

1. the swap term in both Hamiltonian channels (one-center and local-orbital), at full weight;
2. the full Euler–Lagrange potential of the onsite SCAN-L exchange, Eq. (2.18), with the weight w inside the derivatives;
3. exact spin scaling, Eq. (2.20). Evaluating the onsite SCAN-L exchange for the spin-summed density replaces spin-polarized PBE exchange by unpolarized SCAN-L exchange in the swap term and removes a fraction $1 - a_x$ of the onsite exchange splitting; magnetic gaps collapse (NiO 1.0 instead of 2.0 eV);
4. the exact local-orbital projection of Sec. 5.4.

All four were found by checking that the variational optimizer and damped dynamics reach the same state (Sec. 6.7).

5.7 SCAN0r

SCAN0r uses SCAN with the orbital kinetic-energy density everywhere, for the plane-wave density and in the spheres, for exchange and correlation, and replaces a fraction a_x of the onsite SCAN exchange of the local orbitals by exact exchange:

$$E_{\mathrm{xc}}^{SCAN0r} = E_{\mathrm{xc}}^{\mathrm{SCAN}}[n,\tau] + a_x\left(E_x^{\mathrm{HF,1c}} - E_x^{\mathrm{SCAN,1c}}\right). \tag{5.6}$$

It is the meta-GGA analogue of PBE0r, Eq. (5.4), without a swap term, and is selected by **`TYPE='SCAN' TSCANDELOC=T`** with **`!NTBO MODUS='HYBRID' HFWEIGHT=`**a_x. The double counting $E_x^{\mathrm{SCAN,1c}}$ uses the onsite τ of the local orbitals, is spin scaled, and by default contains exchange only (**`TSCANCORR=T`** adds SCAN correlation). Exchange-only spin scaling and the spin-polarized correlation of Chapter 2 apply as for SCAN. With **`TYPE='R2SCAN'`** the same construction gives r^{2}SCAN0r, with r^{2}SCAN everywhere and in the double counting.

5.8 DFT+U

DFT+U is the simplest onsite correction. Instead of exact exchange it adds a Hubbard-like penalty for fractional occupations of one shell of angular momentum ℓ on selected atoms. VESPAW uses the rotationally invariant form of Dudarev *et al.* [55],

$$E_U = \frac{U-J}{2} \sum_R \mathrm{Tr}\big[\mathsf{n}_R - \mathsf{n}_R \mathsf{n}_R\big], \tag{5.7}$$

in which only the difference $U - J$ enters. The penalty vanishes for an occupation matrix with eigenvalues 0 and 1 and is largest for half-filled orbitals, so it favors integer occupations and opens gaps in Mott insulators. The occupation matrix of the shell is obtained from the one-center density matrix,

$$n_{R,ms,m's'} = \sum_{\nu\nu'} Q_{R,\nu\nu'}\, D_{R,(\ell m\nu s),(\ell m'\nu' s')}, \qquad Q_{R,\nu\nu'} = \int_0^{r_U} dr\, r^2\, R_\nu(r) R_{\nu'}(r), \tag{5.8}$$

with the radial parts R_ν of the AE partial waves of the shell, integrated up to the radius r_U (**`RCUT/RCOV`**). This is the occupation matrix of VASP (`LDAUTYPE=2`). Because n is taken as a full spinor matrix, the same expression covers non-spin-polarized, collinear and non-collinear calculations. The correction enters the one-center Hamiltonian,

$$\Delta H_{R,(\ell m\nu s),(\ell m'\nu' s')} \mathrel{+}= Q_{R,\nu\nu'} \frac{\partial E_U}{\partial n_{R,m's',ms}} = Q_{R,\nu\nu'}\,(U-J)\Big[\tfrac{1}{2}\delta_{mm'}\delta_{ss'} - n_{R,ms,m's'}\Big], \tag{5.9}$$

so that DFT+U is part of $\boldsymbol{\theta}$ and is handled by the variational optimizer without special treatment. It is selected per species with **`!STRUCTURE!SPECIES!DFTU L= U[EV]= J[EV]=`**. For NiO with $U - J = 6.3$ eV on the Ni d shell the PBE+U gap is 3.12 eV.

Compared with the local hybrids, DFT+U acts on one shell only, needs a parameter that is not fixed by theory, and its result depends on the definition of the occupation matrix (here the integration radius), so values of U cannot be transferred between codes without care. PBE0r corrects all orbitals, and its single parameter a_x has the meaning of an inverse screening.

Further reading

The original work on the subject of this chapter is found in Refs. [44–48, 55].

Chapter 6

Self-consistency: the variational optimizer

6.1 Introduction . . . 25
6.2 The problem with density mixing . . . 25
6.3 Free energy as a function of the Hamiltonian . . . 26
6.4 The gauge direction . . . 26
6.5 Line search and acceptance . . . 27
6.6 Route 2: direct joint minimization . . . 28
6.7 The consistency test . . . 28
6.8 Certificates of a converged state . . . 29

6.1 Introduction

The Kohn–Sham equations are a fixed-point problem: the density determines the potential, and the potential determines the density. In almost every plane-wave code this problem is solved by mixing input and output densities until they agree, a procedure that works well in most cases and fails, when it fails, without warning. The theory, however, offers a minimum principle. This chapter shows how the self-consistency problem can be written as a minimization in which the effective Hamiltonian is the variable, and every energy is that of an actual state. It describes the structure of this Hamiltonian space, including a gauge mode that is specific to the PAW method, and the line search that turns it into a robust algorithm.

6.2 The problem with density mixing

The standard SCF cycle alternates between solving the Kohn–Sham equations in a fixed potential and building a new input potential by mixing the input and output densities or potentials (Pulay/DIIS [56], Anderson [57], with Kerker preconditioning [58]). The energy along this path is a Harris–Foulkes-type estimate [59, 60]; it is not an upper bound and need not decrease. Mixing can stagnate, oscillate, or converge to an excited state, and nothing in the iteration signals it. VESPAW offers density mixing (**`MIXER='PULAY'`** or **`'ANDERSON'`**) but uses a variational optimizer by default.

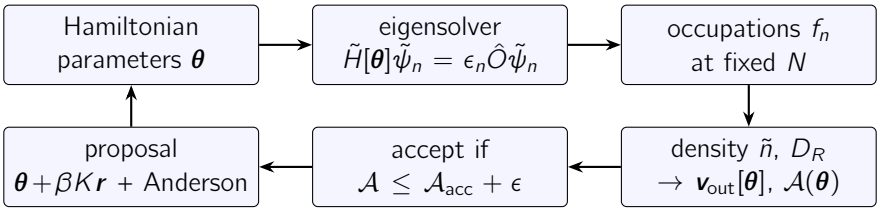


Figure 6.1: One iteration of Route 1. The eigensolver and the occupations turn the trial Hamiltonian $\boldsymbol{\theta}$ into an admissible ensemble, whose free energy $\mathcal{A}(\boldsymbol{\theta})$ is an upper bound of the minimum. The residual $\boldsymbol{r} = \boldsymbol{v}_{\text{out}} - \boldsymbol{\theta}$ is a descent direction. A rejected point is replaced by a backtrack (Sec. 6.5).

6.3 Free energy as a function of the Hamiltonian

We minimize the Mermin free energy, Eq. (2.21), in the PAW form. The parameters of the Hamiltonian are collected in a vector

$$\boldsymbol{\theta} = \big(\tilde{v}(\boldsymbol{r}),\ v_\tau(\boldsymbol{r}),\ \{\Delta H_{R,ij}\}\big). \tag{6.1}$$

For given $\boldsymbol{\theta}$ we determine the eigenstates $\psi_n(\boldsymbol{\theta})$ of $\tilde{H}[\boldsymbol{\theta}]$ and the Fermi occupations $f_n(\boldsymbol{\theta})$ at fixed N, and define

$$\mathcal{A}(\boldsymbol{\theta}) = A\big[\{\psi_n(\boldsymbol{\theta})\}, \{f_n(\boldsymbol{\theta})\}\big]. \tag{6.2}$$

Because $\psi_n(\boldsymbol{\theta})$ and $f_n(\boldsymbol{\theta})$ form an admissible ensemble, $\mathcal{A}(\boldsymbol{\theta})$ is an upper bound of the minimum of A for every $\boldsymbol{\theta}$. At self-consistency, where $\boldsymbol{\theta}$ equals the parameters $\boldsymbol{v}_{\text{out}}[\boldsymbol{\theta}]$ built from the output state, the bound is attained. Figure 6.1 shows one iteration. Following Gonze [61],

$$\frac{\partial \mathcal{A}}{\partial \boldsymbol{\theta}} = \chi(\boldsymbol{\theta})\,\big(\boldsymbol{v}_{\text{out}}[\boldsymbol{\theta}] - \boldsymbol{\theta}\big), \tag{6.3}$$

with χ the static independent-particle response. Since χ is negative semidefinite, the residual $\boldsymbol{r} = \boldsymbol{v}_{\text{out}} - \boldsymbol{\theta}$, and any positive-definite preconditioned version $K\boldsymbol{r}$, is a descent direction; χ never has to be built. The Kerker filter $K(G) = G^2/(G^2 + q_0^2)$ is applied to the plane-wave part of the residual, with q_0^2 set by **`MIXG0SQ`** (default 1.5 bohr^{-2}). The default is chosen for solids. For molecules and slabs with large vacuum it damps the long-wavelength components so strongly that the gains of a step fall below the noise of the iterative eigensolver, and the line search stalls; use $q_0^2 \approx 0.05$–0.1 there.

6.4 The gauge direction

A uniform shift $\tilde{H} \to \tilde{H} + c\hat{O}$ shifts all eigenvalues and the Fermi level by c and leaves orbitals and occupations unchanged. In parameter space it is the direction $\boldsymbol{s} = (1, 0, \{\Delta O_{R,ij}\})$, and $\mathcal{A}(\boldsymbol{\theta} + c\boldsymbol{s}) = \mathcal{A}(\boldsymbol{\theta})$ exactly. The two halves of $\boldsymbol{s}$ are updated by different rules (Kerker on the grid, damped steps in ΔH). If $\boldsymbol{s}$ is not treated explicitly, they drift apart and act as a spurious potential step between spheres and interstitial (Si: convergence to an excited state 28 mH above the ground state). VESPAW therefore

1. projects $\boldsymbol{s}$ out of every residual, and
2. after each step moves $\boldsymbol{\theta}$ along $\boldsymbol{s}$ so that its mean grid potential equals that of the output state.

The second step does not change any energy, but without it $\boldsymbol{\theta}$ keeps its starting gauge while the true potential moves: eigenvalues and Fermi level are then offset (0.49 eV in Si), and with them

the Lagrange multipliers in forces and stress (Chapter 8). In cells with much vacuum the mean potential of the output state can move by hartrees during the iteration. An offset larger than 0.5 H is therefore removed already inside the line search, and the stored eigenvalues are shifted with $\boldsymbol{\theta}$; otherwise the preconditioned eigensolvers start from eigenvalues that lie far off and stall (H_2O layer in an 18 Å cell: eigenvalues 66 eV too low, energy 75 mH too high).

6.5 Line search and acceptance

Proposal. Steps are proposed by an Anderson extrapolation [57] built from *accepted* points only, with a history of **MIXNHISTORY** points (default 4). Each history entry is a pair $(\boldsymbol{\theta}_k, K\boldsymbol{r}_k)$; the proposal is $\boldsymbol{\theta} + \beta K\boldsymbol{r}$, corrected by the Anderson combination of the earlier pairs. The step length β starts at 0.3 and stays between 0.02 and 0.5. It grows by a factor 1.2 after each accepted step, and by 1.5 within a streak of at least three accepted steps. If the Anderson step is more than ten times longer than the plain step $\beta K\boldsymbol{r}$, the plain step is used instead, because it is a descent direction.

Acceptance. A trial point $\boldsymbol{\theta}'$ is accepted if

$$\mathcal{A}(\boldsymbol{\theta}') \leq \mathcal{A}_{\mathrm{acc}} + \epsilon, \qquad \epsilon = 10^{-7}\ \mathrm{H}, \tag{6.4}$$

where $\mathcal{A}_{\mathrm{acc}}$ is the energy of the *last accepted* point. The slack ϵ absorbs the noise of the iterative eigensolver. It is measured from the last accepted energy and not from the lowest one: with the lowest energy as reference, a single evaluation that came out low by chance blocked all later steps of noisy runs (H_2O in vacuum stalled at a residual of 10^{-3}). The price is that a stalled run can creep upward by up to ϵ per accepted step, so the accepted energies are monotone only within ϵ.

Rejection. A rejected trial point is not added to the history. The response depends on the number of consecutive rejections:

1. after the first, the next trial is the midpoint between the rejected point and the last accepted point;
2. after the second, the Anderson history is discarded, β is halved, and the plain damped step $\beta K\boldsymbol{r}$ is taken from the last accepted point;
3. further rejections halve the step towards the last accepted point;
4. after five consecutive rejections, the last accepted point is evaluated again and accepted whatever its energy ("re-anchoring"), because its stored energy may stem from eigenstates that were less converged than the present ones.

β is halved once per rejection episode, not at every rejection; halving it at every backtracking step made it collapse to its lower bound, from which it needed some 20 accepted steps to recover (fcc Al: 77 instead of 14 iterations). A rejection right after a single accepted backtracking step also halves β. Without this rule the iteration can fall into a period-2 cycle, in which the full step always overshoots and its half is always accepted (NiO: half of the trial points rejected).

Convergence. The run stops when the mean-square grid residual of an accepted point is below **MIXTOL** and the largest one-center residual (of ΔH and v_τ) below its square root; the grid alone can converge while ΔH still lags, leaving the energy tens of microhartree high. It also stops after two consecutive accepted steps that each lowered the energy by less than **EDIFF**, provided that both were full-length steps ($\beta \geq 0.25$) not preceded by a rejection, because after backtracking a

Algorithm 1 Hamiltonian-space variational SCF (Route 1).

$\boldsymbol{\theta}_0 \leftarrow \boldsymbol{v}_{\text{out}}[\psi_{\text{start}}]$; hold fixed for a short eigensolver warm-up
loop
  diagonalize $\tilde{H}[\boldsymbol{\theta}]$ (Davidson, RMM-DIIS or CG); Fermi occupations at fixed N
  evaluate $\mathcal{A}(\boldsymbol{\theta})$ at the output state; build $\boldsymbol{v}_{\text{out}}$
  $\boldsymbol{r} \leftarrow \boldsymbol{v}_{\text{out}} - \boldsymbol{\theta}$; project out $\boldsymbol{s}$
  if $\mathcal{A}(\boldsymbol{\theta}) \leq \mathcal{A}_{\text{acc}} + \epsilon$ **then**
    accept: $\mathcal{A}_{\text{acc}} \leftarrow \mathcal{A}(\boldsymbol{\theta})$; add $(\boldsymbol{\theta}, K\boldsymbol{r})$ to the history; stop if converged
    $\boldsymbol{\theta} \leftarrow$ Anderson proposal with step length β
  else
    backtrack towards the last accepted point; drop the history after two rejections
    re-evaluate the last accepted point after five rejections
  end if
  re-gauge $\boldsymbol{\theta}$ along $\boldsymbol{s}$ to the mean grid potential of $\boldsymbol{v}_{\text{out}}$
end loop

small gain only reflects a short step. Neither criterion proves that the state is the minimum; the certificates of Sec. 6.8 test that.

Route 1 is selected by **`!PSIDYN MIXER='VARIATIONAL'`** together with **`OPTIMIZER='DAV'`**, **`'RMM'`** or **`'CG'`**; it is the default mixer.

Hybrid operators. The onsite hybrid operator of Chapter 5 depends on the orbitals through D, not only on the density. It is taken from the current state rather than line-searched; including its matrix elements in $\boldsymbol{\theta}$ left a residual that did not decay.

6.6 Route 2: direct joint minimization

As an alternative (**`OPTIMIZER='EDFT'`**), A is minimized directly with respect to the orbitals and an auxiliary Hamiltonian H^{aux} whose eigenvalues define the occupations, in the spirit of ensemble DFT [62, 63]. The orbital direction is the Teter-preconditioned [64] projected residual combined by Polak–Ribière conjugate gradients; the direction for H^{aux} is $H^{\text{phys}} - H^{\text{aux}}$. Trial points are built by Löwdin orthonormalization and rotated into the eigenbasis of H^{aux}; step lengths come from a quadratic model, with the same acceptance and re-anchoring logic as Route 1. The orbital direction is computed from a Kerker-preconditioned Hamiltonian. With this preconditioning Route 2 currently stops about 5×10^{-6} H above the minimum on Si; without it (`VESPAW_EDFTVAR_NOKERKER=1`) it reaches the minimum.

6.7 The consistency test

The variational optimizer minimizes the energy. Damped Car–Parrinello dynamics (Chapter 7) and density mixing converge to a self-consistent state. The two coincide only if the Hamiltonian is the derivative of the energy. A missing potential term often shows up as the variational optimizer finding energies *below* the self-consistent state without converging, or as many rejected steps. The regression suite (Chapter 9) therefore compares optimizers in every case.

The test is sensitive but not sufficient. When the Hamiltonian deviates from the derivative only a little, all optimizers follow the same wrong Hamiltonian and agree with each other. In MgO with SCAN0r, a double-counting term that subtracted the potential of the core kinetic-energy density (Chapter 2) left the energy correct but the Hamiltonian inconsistent. Pulay and Anderson mixing, Route 1 with Davidson and RMM-DIIS, and friction dynamics all reported convergence

and agreed within 0.25 meV, at a state 7.5 meV above the minimum with a band gap 42 meV too small. Route 1 rejected only one to nine steps. For a charged cell and for NiO the signal was strong, with more than a third of the steps rejected. The direct test is the Janak certificate of the next section.

6.8 Certificates of a converged state

A small residual and a small change of the energy are necessary conditions for convergence, but they are not sufficient. They are also fulfilled by a calculation that has stalled, and by the self-consistent solution of a Hamiltonian that is not the derivative of the reported energy. The variational formulation provides two tests of a final state, which can be applied to the result of any optimizer, and of any code.

C1: descent probe. The final state is used as the start of Route 1. Because Route 1 accepts only steps that do not raise $\mathcal{A}$ (within ϵ), a decrease of the energy by more than the tolerance proves that the state was not stationary, and it yields a lower upper bound. If the first accepted steps change $\mathcal{A}$ only within the tolerance, the state is stationary with respect to $\boldsymbol{\theta}$ for the Hamiltonian that the code constructs. In VESPAW the probe is a restart (**`START=F`**) with **`OPTIMIZER='DAV' MIXER='VARIATIONAL'`**.

C2: frozen-orbital Janak test. If the Hamiltonian is the derivative of the energy, the derivative of the energy with respect to the occupation f_n of an eigenstate, at fixed orbitals, equals its eigenvalue [65],

$$\Delta_n = \frac{\partial \mathcal{A}}{\partial f_n}\Big|_{\psi} - \epsilon_n = \langle \psi_n | \hat{H}_{\mathcal{A}} - \tilde{H}[\boldsymbol{\theta}] | \psi_n \rangle \tag{6.5}$$

must vanish, where $\hat{H}_{\mathcal{A}}$ is the functional derivative of the implemented energy and $\tilde{H}[\boldsymbol{\theta}]$ the Hamiltonian whose eigenvalues are reported. The derivative is evaluated for the lowest unoccupied and the highest occupied state by the second-order one-sided difference

$$\frac{\partial \mathcal{A}}{\partial f_n} \approx \frac{-3\mathcal{A}(0) + 4\mathcal{A}(q) - \mathcal{A}(2q)}{2q}, \tag{6.6}$$

with an occupation change $q = +0.01$ for the lowest unoccupied and $q = -0.01$ for the highest occupied state. The energies are those of the *frozen* orbitals, without self-consistency. In VESPAW each energy is one step of a restart with the electron number changed by **`!STRUCTURE!OCCUPATIONS CHARGE[E]`**, **`!CONTROL!GENERIC NSTEP=1`** and **`!PSIDYN STOP=T`**. The test costs four energy evaluations. For a Mermin ensemble the derivative of the entropy term has to be subtracted.

What they detect. The two tests are complementary.

- C1 detects a state that is not stationary, but it inherits an error of the Hamiltonian, because the probe uses the same Hamiltonian. In the MgO example above it passed ($|\Delta\mathcal{A}| \leq$ 0.23 meV).
- C2 detects an inconsistency between energy and Hamiltonian. In the MgO example it failed by -92 and -190 meV for the lowest unoccupied and the highest occupied state, against about 1 meV for the correct code. It does not detect a state that is unconverged only in parts that do not affect the frontier states. A density mixer whose residual included the corner components of the FFT box, which the Kerker filter never mixes, stopped 29 meV above the minimum with an unrelaxed one-center density matrix; C2 passed and C1 lowered the energy by 29.45 meV.

- Neither test detects a functional that is implemented incorrectly but consistently, together with its derivative. The factor-two error in the one-center kinetic-energy density (Chapter 2) passed both; it was found by checking $\int \tau$ against the kinetic energy and by comparing band gaps with another code.

The certificates thus establish that a state is a stationary point of the implemented functional, not that the functional is the intended one, nor that the stationary point is the global minimum. They are not specific to VESPAW: applied to VASP, C1 showed that antiferromagnetic NiO with PBE+U and the default `LMAXMIX=2` stops 56 μeV above the minimum, while C2 passed within 1 meV.

Further reading

The original work on the subject of this chapter is found in Refs. [61–63, 65, 66].

Chapter 7

Eigensolvers and dynamics

7.1 Introduction 31
7.2 Block Davidson 31
7.3 RMM-DIIS 32
7.4 Conjugate gradients 32
7.5 Car–Parrinello dynamics with friction 32

7.1 Introduction

Most of the computing time of a calculation is spent in finding the lowest eigenstates of the Hamiltonian. For a few hundred bands in a basis of tens of thousands of plane waves, a direct diagonalization is out of the question. Iterative methods instead improve a set of trial wave functions, using only the action of the Hamiltonian on them. In the original Car–Parrinello approach this is done by a fictitious dynamics of the wave functions. This chapter describes the iterative eigensolvers of VESPAW and the damped dynamics of CP-PAW, and what each of them requires to return reliable orthonormal eigenstates.

For Route 1 the eigenstates of $\tilde{H}[\boldsymbol{\theta}]$ must be found at every iteration. They are warm-started from the previous iteration and converged iteratively. All solvers work with the generalized problem $\tilde{H}\psi = \epsilon\hat{O}\psi$ and apply $\tilde{H}$ and $\hat{O}$ in blocks of bands.

7.2 Block Davidson

`OPTIMIZER='DAV'`. Each band is extended by a preconditioned correction vector $(\tilde{H} - \epsilon\hat{O})\psi$; the new vectors are projected out of *all* current bands (twice), vectors lost to cancellation are dropped, and the extended basis is canonically orthogonalized with an eigenvalue cutoff of 10^{-10}. A Rayleigh–Ritz step in this subspace gives the new bands. Converged bands are locked. The implementation keeps seven wave-function-sized arrays. Davidson is robust from any start and is the reference solver.

Pitfall. An earlier Davidson orthogonalized only against converged bands and stabilized the near-singular overlap with a shift; the returned orbitals were not orthonormal, which produced energies below the minimum and split degenerate states.

7.3 RMM-DIIS

OPTIMIZER='RMM' . Residual minimization with direct inversion in the iterative subspace [67, 68] minimizes the norm of the residual of each band separately, without orthogonalization against other bands, and is therefore cheap and parallel. It converges to the eigenvector nearest the starting guess. Started from random orbitals it converges to wrong states, so the first **RMMDELAY** iterations (default 8) use Davidson. Bands are processed in blocks of 32 (**NRMMX** inner sweeps). RMM-DIIS is the fastest option for large systems (Si_{64}: 247 s vs 386 s for Davidson).

Pitfall. With the onsite hybrid, RMM-DIIS reaches the correct energy but can miss the lowest empty band (MgO PBE-SL0r gap 6.54 instead of 4.23 eV). Take band gaps of hybrid calculations from Davidson.

7.4 Conjugate gradients

OPTIMIZER='CG' . Band-by-band preconditioned conjugate-gradient minimization of the Rayleigh quotient with orthogonalization to the lower bands, followed by a subspace rotation. It is reliable, but slower than the block methods for many bands.

7.5 Car–Parrinello dynamics with friction

The original CP-PAW optimizer (**OPTIMIZER='MD'** , the default when no eigensolver is chosen) treats the wave-function coefficients as classical degrees of freedom with a fictitious mass m_ψ [69]. Instead of solving the electronic problem at each atomic configuration, as in Born–Oppenheimer dynamics, Car and Parrinello propagate orbitals and nuclei together with the Lagrangian

$$\mathcal{L} = \sum_n f_n\, m_\psi \langle \dot{\tilde\psi}_n | \dot{\tilde\psi}_n \rangle + \sum_I \tfrac{1}{2} M_I \dot{\boldsymbol{R}}_I^2 - E\big[\{\tilde\psi_n\}, \{\boldsymbol{R}_I\}\big] + \sum_{nm} \Lambda_{mn} \big(\langle \tilde\psi_n | \hat{O} | \tilde\psi_m \rangle - \delta_{nm} \big). \tag{7.1}$$

The terms are the fictitious kinetic energy of the orbitals, the kinetic energy of the nuclei, the total energy, and the orthonormality constraint with Lagrange multipliers Λ. The Euler–Lagrange equations give Newton's equations for the nuclei, with the forces of Chapter 8, and, after division by f_n and absorbing the occupations into Λ, the equation of motion of the orbitals with an added friction term,

$$m_\psi \ddot{\tilde\psi}_n = -\tilde{H}\tilde\psi_n + \sum_m \hat{O}\tilde\psi_m \Lambda_{mn} - \gamma\, m_\psi \dot{\tilde\psi}_n, \tag{7.2}$$

with a friction γ (**!PSIDYN FRIC**). With friction the dynamics relaxes into the ground state. The "autopilot" (**!PSIDYN!AUTO**) lowers the friction when the energy decreases and raises it when it increases. Damped dynamics is slow but very robust, converges to the self-consistent state for any functional, and serves as the reference in the consistency test. **SAFEORTHO=T** orthogonalizes such that the energy is conserved; it should not be used with variable occupations.

Further reading

The original work on the subject of this chapter is found in Refs. [33, 67–70].

Chapter 8

Forces, stress and molecular dynamics

8.1 Introduction 33
8.2 Forces 33
8.3 Stress 34
8.4 Structure relaxation and molecular dynamics 34

8.1 Introduction

Once the electrons are in their ground state, the atoms feel forces and the cell feels a stress. These derivatives of the energy are what makes structure optimization and molecular dynamics possible. In a variational theory they follow from the explicit dependence of the energy on the atomic positions, but in the PAW method this dependence hides in many places: the projector functions, the compensation charges and the overlap operator all move with the atoms. This chapter collects the expressions for forces and stress and describes the equations of motion for the atoms.

8.2 Forces

The force on atom R is the total derivative of the energy with respect to its position. At a stationary point of the energy the orbital response drops out (Hellmann–Feynman [4]). In PAW the projectors, the compensation charges and the one-center terms move with the atom, and the orthonormality constraint depends on the positions through $\hat{O}$. The derivative of the Lagrangian $E - \sum_{nm} \Lambda_{mn} \big(\langle \tilde{\psi}_n | \hat{O} | \tilde{\psi}_m \rangle - \delta_{nm} \big)$ at fixed $\tilde{\psi}_n$ gives

$$\begin{aligned} \boldsymbol{F}_R = - \int d^3 r \left[\tilde{v}_{\mathrm{H}}(\boldsymbol{r}) \frac{\partial \hat{n}_R(\boldsymbol{r})}{\partial \boldsymbol{R}} + \tilde{\mu}_{\mathrm{xc}}(\boldsymbol{r}) \frac{\partial \tilde{n}^{\mathrm{c}}_R(\boldsymbol{r})}{\partial \boldsymbol{R}} + \tilde{n}(\boldsymbol{r}) \frac{\partial \bar{v}_R(\boldsymbol{r})}{\partial \boldsymbol{R}} \right] \\ - 2 \,\mathrm{Re} \sum_n \sum_{ij} \langle \tilde{\psi}_n | \frac{\partial \tilde{p}_{R,i}}{\partial \boldsymbol{R}} \Big\rangle \Big[\Delta H_{R,ij} \, f_n \langle \tilde{p}_{R,j} | \tilde{\psi}_n \rangle - \Delta O_{R,ij} \sum_m \langle \tilde{p}_{R,j} | \tilde{\psi}_m \rangle \Lambda_{mn} \Big]. \end{aligned} \tag{8.1}$$

The first line is the force of the plane-wave potentials on the charges that move with the atom: the electrostatic potential $\tilde{v}_{\mathrm{H}}$ of $\tilde{n} + \hat{n}$ on the compensation charge $\hat{n}_R$ (which, in CP-PAW, also carries the nuclear and core charge, so the ion–ion interaction is included), the exchange-correlation potential $\tilde{\mu}_{\mathrm{xc}}$ on the pseudized core density $\tilde{n}^{\mathrm{c}}_R$, and the local potential $\bar{v}_R$ on the pseudo density. The compensation charges of neighboring atoms overlap; their direct interaction, which the plane-wave grid cannot resolve, is added analytically. The second line is the projector

term. The one-center energies depend on the atom only through the density matrices D_R, so moving the projectors changes the energy by ΔH times the change of D_R, and the overlap constraint by ΔO times the change of $\langle\tilde\psi_n|\hat O|\tilde\psi_m\rangle$. The multipliers contain the occupations, $\Lambda_{mn} = \frac{1}{2}(f_m + f_n)\langle\tilde\psi_m|\tilde H|\tilde\psi_n\rangle$. In Car–Parrinello dynamics Λ comes from the orthonormality constraint. For the eigensolvers the orbitals are Ritz vectors, and $\Lambda_{mn} = f_n\epsilon_n\delta_{mn}$. With non-collinear spinors, ΔH and the projections carry spin indices, and the sum runs over them as well.

Pitfalls. Before September 2026 the eigensolvers did not set Λ at all, so the constraint term was missing (0.5% error in the Si force), and with the variational mixer the eigenvalues carried the gauge offset of the previous chapter. Both are now fixed, and forces agree with finite differences of the energy to 10^{-4} relative (regression case `si_forces`).

8.3 Stress

The stress tensor is the derivative of the energy with respect to a homogeneous strain ϵ,

$$\sigma_{\alpha\beta} = -\frac{1}{V}\frac{\partial E}{\partial \epsilon_{\alpha\beta}}, \tag{8.2}$$

following Nielsen and Martin [71]; positive values mean that the cell wants to expand, and the pressure is $\operatorname{tr}\sigma/3$. It contains kinetic, Hartree, exchange-correlation, projector and compensation-charge terms, and the same constraint term as the forces.

The analytic stress is taken at a fixed set of plane waves (the basis strains with the cell). A finite difference at fixed cutoff changes the basis and contains the *Pulay stress*, which vanishes as the cutoff grows (Si: 2% at 25 Ry, 0.5% at 50 Ry). Lattice constants from energy-volume curves need converged cutoffs for the same reason.

Output. **`!CONTROL!GENERIC FORCES=T STRESS=T`** evaluates forces and stress also for fixed atoms and a fixed cell and prints them at full precision at the end of the run (`FORCE`, `STRESS`, `PRESSURE[GPA]` lines).

8.4 Structure relaxation and molecular dynamics

Atomic motion is controlled by **`!CONTROL!RDYN`**. With friction (**`FRIC`**, **`!RDYN!AUTO`**) the atoms relax into a minimum; without friction they follow Newton's equations. The cell shape can be relaxed or propagated with **`!CONTROL!CELL MOVE=T`** [71].

L-BFGS relaxation. With **`!RDYN OPTIMIZER='LBFGS'`** the geometry is optimized by the limited-memory BFGS method [72, 73], a quasi-Newton method that builds an approximate inverse Hessian from the last **`NMEM`** (default 10) pairs of steps and force changes. The electrons are converged at each geometry with an eigensolver and Route 1 or a density mixer, so this is a Born–Oppenheimer optimization. A moving cell is optimized together with the atoms. The variables are

$$\boldsymbol{x} = \big(\boldsymbol{u}_1, \ldots, \boldsymbol{u}_M,\ L(\mathsf{F} - \mathbf{1})\big), \qquad \boldsymbol{u}_I = \mathsf{F}^{-1}\boldsymbol{R}_I, \tag{8.3}$$

where $\mathsf{F} = \mathsf{T}\mathsf{T}_0^{-1}$ is the deformation of the cell T relative to the reference cell T_0 of the input, and $\boldsymbol{u}_I$ are the undeformed positions. The generalized forces are $\mathsf{F}^{\mathsf{T}}\boldsymbol{F}_I$ for the atoms and $[\mathsf{S} - V(P\mathbf{1} + \boldsymbol{\sigma}_{\text{ext}})]\mathsf{F}^{-\mathsf{T}}/L$ for the cell, with the internal stress $\mathsf{S} = -\partial E/\partial\epsilon$ and a target pressure P and stress $\boldsymbol{\sigma}_{\text{ext}}$ (**`!CELL P[GPA]`**, **`STRESS[GPA]`**). For a pure pressure they are the gradient of the enthalpy $H = E + PV$. The target acts on the actual cell (a true, Cauchy stress), so the

result does not depend on the starting cell. The length $L = \sqrt{3B_0V_0/h}$, with a bulk modulus B_0 and the curvature h of the initial Hessian, makes the first step of a hydrostatic strain exact and balances atom and cell steps.

The method requires accurate energies. Near the minimum an L-BFGS step lowers the energy only by about $F^2/2h$, which for small forces is comparable to the default SCF tolerance; the energy test and the curvature pairs then turn into noise. After each step **EDIFF** and **MIXTOL** are therefore tightened to 1% of the predicted energy decrease, but not below 10^{-9} H. Four safeguards make the optimization robust:

1. a step that raises the energy (enthalpy) by more than the noise is halved back towards the last accepted point;
2. a pair without positive curvature is not stored, and the history is cleared;
3. a direction that is not downhill is replaced by steepest descent;
4. a trust radius limits the displacement of every atom, starting at **MAXSTEP[AA]**, halved at every backtrack and doubled after a successful step.

The optimization stops when the largest force is below **FMAX[EV/AA]** and, with a moving cell, the largest deviation of the stress from its target below **!CELL SMAX[GPA]**.

Pitfall. For molecules and slabs in vacuum the variational mixer leaves an energy error of about 10^{-5} H that depends on the geometry. This stalls L-BFGS near 0.05 eV/Å; use **MIXER='PULAY'** there. A cell relaxation needs a higher cutoff than the energy, because the stress converges more slowly with the cutoff (Pulay stress, Sec. 8.3): for $LiMn_2O_4$ with PBE0r the pressure at a fixed cell was -17.1, -2.3 and -1.5 GPa at 50, 65 and 80 Ry.

Car–Parrinello vs Born–Oppenheimer MD. In Car–Parrinello MD the wave functions are propagated together with the atoms and stay close to the ground state if their fictitious mass is small enough (adiabaticity [74]). The time step must resolve the fastest wave-function motion. In Born–Oppenheimer MD the electrons are converged at each step (with an eigensolver and Route 1); the time step is set by the ions, but each step costs a full SCF.

Thermostats. Temperature is controlled by Nosé–Hoover thermostats [75] on the atoms (**!RDYN!THERMOSTAT[K]**) and, in Car–Parrinello MD, on the wave functions [76] (**!PSIDYN!THERMOSTAT**), which removes the heat that leaks from the ions into the fictitious electronic motion.

Further reading

The original work on the subject of this chapter is found in Refs. [71, 73–76].

Chapter 9

Structure of the code

9.1 Introduction 36
9.2 Source files 37
9.3 One SCF step 37
9.4 Parallelization 37
9.5 Input 38
9.6 Adding a functional 38
9.7 Adding an optimizer 38
9.8 Regression suite 39

9.1 Introduction

This chapter is written for those who want to read the source code, change it, or add to it. It gives an overview of the source files, follows one self-consistency step through the program, and describes how a new functional or a new optimizer is added and tested.

9.2 Source files

File	Content
`vespaw.f90`, `vespaw_driver.f90`	main program, time-step loop, reports
`vespaw_ioroutines.f90`	reading `.cntl/.strc` input
`vespaw_waves1.f90`, `vespaw_waves2.f90`	wave functions, density, $H\psi$, forces, total energy
`vespaw_cg.f90`	Davidson, RMM-DIIS, CG
`vespaw_mixer.f90`	variational, Pulay and Anderson mixers
`vespaw_edft.f90`	Route 2 (EDFT)
`vespaw_potential.f90`	plane-wave potential, electrostatics, XC on the grid
`vespaw_augmentation.f90`	one-center (PAW sphere) terms
`vespaw_setups.f90`	setup construction and tests
`vespaw_lmto.f90`	local orbitals, PBE0r, PBE-SL0r, SCAN0r
`vespaw_scantau.f90`, `vespaw_scannsph.f90`	SCAN with orbital τ, non-spherical one-center SCAN
`vespaw_dft.f90`	exchange-correlation functionals
`vespaw_atoms.f90`, `vespaw_cell.f90`	ionic and cell dynamics
`vespaw_occupations.f90`	occupations, Mermin
`vespaw_onsite.f90`	DFT+U, spin–orbit coupling
`vespaw_kpoints.f90`	k-point mesh, space-group symmetry
`vespaw_schroedinger.f90`	radial (scalar-relativistic) equations

The code is organized in CP-PAW's object style: each module keeps its data in a Fortran module and is accessed through `OBJECT$SET`, `$GET` and action routines.

9.3 One SCF step

1. `PAW` (driver) calls `TIMESTEP`, which calls `WAVES$ETOT`.
2. `WAVES$ETOT`: projections $\langle\tilde{p}|\tilde{\psi}\rangle$, density and one-center density matrices; plane-wave potential (`WAVES_VOFRHO`); one-center potentials and ΔH (`AUGMENTATION`); hybrid terms (`LMTO`); forces and stress (`WAVES$FORCE`).
3. Variational mixer (`WAVES_VARMIXHAMILTONIAN`): the output Hamiltonian and the energy are handed to the line search, which returns $\boldsymbol{\theta}$ for the next diagonalization.
4. Eigensolver (`DAVIDSON$STATE_BY_BLOCK`, `RMM$STATE_BY_BLOCK`, `CG$STATE_BY_STATE`) or, for dynamics, $H\psi$ and propagation.
5. Reports (`PRINFO`), capture of forces and stress, and the switch of all dynamical variables to the next step.

9.4 Parallelization

VESPAW inherits the MPI parallelization of CP-PAW. The tasks are divided into k-point groups. Each k-point, with all its bands and spins, belongs to one group, and the points are distributed so that the groups have similar work (a point with super wave functions counts half). Inside a group the plane-wave coefficients and the real-space grid are split into sheets of planes along the first lattice vector, and the fast Fourier transforms are distributed over the tasks of the group. The density and the plane-wave potential live on a second layout, in which the grid is split over

all tasks. Densities and one-center density matrices are summed over the k-groups; forces and stress over all tasks.

The methods added in VESPAW were written for one task and had to be made consistent with this layout:

- the eigensolvers loop over the local k-points of their group, and the sums in their subspace matrices and projections run over the tasks of that group;
- the variational mixer, the Pulay and Anderson mixers and Route 2 work on the density layout: the potential is moved from the wave-function layout, mixed, and moved back, and every scalar product of grid functions is summed over all tasks;
- the symmetrization of the density assembles the full grid on every task, symmetrizes it, and keeps the local sheets.

A run on several tasks must reproduce the serial run. This holds to 10^{-9} H or better for Si, Al, NiO with U and with the hybrids, spin–orbit coupling, the dipole slab, symmetry and the Si_8 Γ case. The regression suite runs under MPI with `VESPAW_REGRESSION_NP=4`.

The MPI executable is built with `make fast_parallel_new` (`vespaw_fast_mpi.x`). Linear algebra and the one-center SCAN terms also use threads. With MPI, set `OMP_NUM_THREADS=1` and `OPENBLAS_NUM_THREADS=1`, unless there are more cores than tasks.

9.5 Input

The `.cntl` and `.strc` files are read into linked lists (`LINKEDLIST$...`). Blocks start with `!NAME` and end with `!END`; a block whose name is changed (e.g. `!ORBPOT_OFF`) is ignored. `READ=F` inside a block also switches the block off. The routines `READIN_*` in `vespaw_ioroutines.f90` transfer the values to the modules.

9.6 Adding a functional

1. Implement energy density and potential in `vespaw_dft.f90` (or a module of its own), register a type name in `READIN_DFTTYPEFROMNAME`.
2. Provide the one-center evaluation (`vespaw_augmentation.f90`), with the full potential, Eq. (2.18).
3. Verify: energy vs finite difference of the potential along a density change; variational optimizer vs damped dynamics (Sec. 6.7); forces and stress vs finite differences.
4. Add a regression case.

9.7 Adding an optimizer

Optimizers are dispatched on `OPTIMIZERTYPE` in `WAVES$ETOT`. A new eigensolver must return orthonormal Ritz vectors and their Ritz values in `THIS%EIGVAL` (used for the Lagrange multipliers of the forces). It should be tested against Davidson on the whole regression suite.

9.8 Regression suite

`regression/run.py` runs all cases (`regression/*/case.json`) or selected ones (`run.py CASE:RUN`). A case has template inputs with `@PLACEHOLDER@` substitutions, several runs (optimizers, cold and seeded starts, multi-stage runs sharing a restart file), checks against reference values, agreement tests between runs, finite-difference tests of forces and stress, and a list of known issues. The cases cover PBE Si (all optimizers), hot Al (metal, Mermin), Si_8 at Γ, NiO AFM (PBE, PBE+U, PBE0r, PBE-SL0r, SCAN0r, TSCAN), MgO PBE-SL0r, TSCAN Si, spin–orbit Si, a polar slab with dipole correction, space-group symmetry, and forces and stress. Every change to the code should leave the suite passing.

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